



KENYATTA UNIVERSITY
INSTITUTE OF OPEN AND VIRTUAL LEARNING
GEOGRAPHY DEPARTMENT
AGE 110: INTRODUCTION TO STATISTICS, MAP ANALYSIS AND
CARTOGRAPHY

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PART ONE

SECTION 1

1.0 Introduction to Statistics in Geography

1.1 Introduction

This is an introductory Section to statistics, which provides the definition and types of statistics. It also gives the purpose of statistical analyses, especially the meaning of statistical significance in geography. It demonstrates that even proper use of statistics does

not guarantee useful results. Other important aspects introduced in this Section are a number of new terms, concepts and symbols that are used quite often in statistics.

1.2 The objectives

- (i) Provide the meaning and types of the word statistics.
- (ii) Discuss the role of statistics in geographical studies.
- (iii) To introduce the basic terminologies and some notations, which are commonly encountered in the study of statistics.

1.3 Definition

Webster's New Dictionary gives two definitions of the word statistics:

- a) Facts or data of a numerical kind, assembled, classified and tabulated so as to present significant information about a given subject.
- b) The science of assembling, classifying and tabulating such facts or data.

However, statistics entails much more than what the above definitions can convey. Statisticians assemble, classify and tabulate data. They also analyze data in order to make generalizations and decisions (Weiss 1996). For example, a city council may decide where to locate a dumping site based on environmental impact assessment statements, which include a variety of statistical data.

According to Levin (1978) statistics is the application of methods used in collecting, processing, summarizing and analyzing data/figures. The data/figures, which result from the statistical analysis and summary such as the mean, mode, median, and standard deviation, among others are also referred to as statistics.

1.4 Types of Statistics

In Geography, statistical techniques are applied in human, physical, and environmental areas of the discipline. The techniques are so varied that geographers usually separate them into two major categories namely:

- a) Descriptive statistics
- b) Inferential statistics.

Descriptive Statistics consists of methods for organizing and summarizing information from a population. It includes the construction of charts, graphs and tables, and computation of various descriptive measures such as averages, measures of dispersion and percentiles.

Inferential statistics consists of methods for drawing and measuring the reliability of conclusions about a population based on information obtained from a sample of the population. Polling opinion of political voting provides a good example of inferential statistics. It would be expensive and unrealistic to interview all Kenyans on their voting preferences in a coming General Election. Statisticians who want to gauge the sentiment of the entire Kenyan voters on any polling project can afford to interview only a carefully chosen group of a few thousand, or tens of thousand voters. This group is referred to as a sample of the Kenyan voters. Then the statisticians can analyze the information obtained from the sample of the voting population to draw conclusions on the preferences of the entire voting population. Inferential statistics thus provides methods for making such inferences.

1.5 Significance of statistical Methods in Geography

- i) Statistical methods enable geographers to handle large quantities of numeric data and summarize them in a form that can easily be described and explained. Geographers like many other scientists are to deal with „information explosion“. In order to make effective use of this huge information capacity, the geographer needs ways of summarizing large datasets, so that concise measures of their characteristics are obtained.
- ii) Statistical methods enable geographers to establish precisely the form, nature and degree of relationships between spatially covariant phenomena. The methods are concerned with processes that are in operation, and which always engender identical results in identical circumstances. For instance, armed with this kind of information geographers should be able to predict the relationship between temperature and altitude of a certain location. It should, however, be noted that very few geographical processes are of deterministic nature. As noted by Edmon (1978) they frequently

behave in different ways at different times, and one can hardly predict with certainty of the outcome of a process even under carefully controlled conditions.

- iii) The techniques minimize subjective judgment, and thus increasing objectivity and precision in explaining the spatial trends of geographical distributions and relationships. Note that most of the geographical data is obtained from **samples**, rather than with all data from the population. In such a case it is assumed that the sample is a representative of the total dataset of the population or the universe from which it has been drawn. This is where inferential statistics is applied with certain defined limits, that one can make statements about the characteristics of a population based only on data collected from a sample (refer to voting preferences under types of statistics above).
- iv) Statistical significance is one of the most powerful uses of statistics, in helping to decide whether an observed relationship between two sets of sample data is statistically significant. Statistical significance is concerned with whether an observed difference, for instance, between two samples can be taken as signifying something else. It helps to establish if there is a difference within the population from which the samples are drawn. In other words, is the difference appearing in the samples merely as a result of chance in the sampling procedure?
- v) The statistical techniques help geographers to relate geographical studies to other scientific disciplines. For instance, the use of hypotheses to be tested by statistical methods encourages geographers to have original thoughts (Ogonda 1991). It also results in contributions to a wider community of interest for the purpose of planning and development. Reasoning and judgment are exercised and knowledge gained from one situation can be applied to another discipline of different nature and similar results obtained. Thus, holding an important position in connecting geographical models or theories/concepts with the physical and human environment in which we live.

1.6 Misuse of Statistics

The misuses are related to errors in the use of the basic definitions underlying an investigation, in the application of those definitions in the measurement or classification of individual objects or people, and in the selection of individuals

for measurement. They also illustrate errors in the use of the resulting data by making improper comparisons, by failing to allow for such indirect causes of differences as heterogeneity of groups with regard to important variables, by disregarding the variability that is usually present even under apparently constant conditions, by technical errors, and by misleading verbal or graphical presentations.

1.7 Key Words and Symbols

- a) **A parameter.** This is a description of a particular characteristic of the distribution of a variable, and therefore provides a description of the population itself.
- b) **Population.** This is a data set representing the entire entity of interest. In other words, it refers to the totality or universe of observations. For instance, a clearly defined group of cases such as all people in Kenya forms a population.
- c) **Sample.** This is a limited number of observations, individuals or cases drawn from the population such as 10% of Kenyan population.
- d) **Variable.** It is a characteristic that is being measured or a measurable property of any person, place, or other phenomena of interest. Examples of variables for human beings are height, weight, a number of siblings, sex, marital status and eye color. The first three of these variables yield numerical information and are example of quantitative variables; the last three yield non-numerical information, and are examples of qualitative variables.

Symbols

Σ = The symbol Σ (the upper-case version of the Greek letter sigma) implies the operation of summing. If x stands for a variable, then Σx means “sum all the

observations” of x .

$\bar{X}$ = The sample mean, obtained by summing all the observed values of x and dividing by the number of observed values. The formula for this is $\bar{X} = \Sigma x/n$,

Where n is the number of observed values

We pronounce $\bar{X}$ as “ x bar”.

1.8 Summary

- Statistics has been defined as numerical facts pertaining to a body of objects or people, or as a science that deals with methods of data collection, analysis and drawing of inferences
- It is possible to perform a descriptive study on a sample as well as on a population. It is only when an inference is made about the population based on information from the sample does the study become inferential. In other words, descriptive and inferential statistics are somewhat interrelated.
- It is almost always necessary to employ techniques of descriptive statistics to organize and summarize the information from the sample before carrying out any inferential analysis. Moreover, the preliminary descriptive analysis of a sample often reveals features that lead to the choice of, or reconsideration of the choice of the appropriate inferential method.

1.9 Exercise

- (1) Using relevant illustration, explain the relationships between a sample and population.
- (2) Define the following terms as concisely as possible:
 - a) (i) Statistics, (iii) Inferential Statistics.
 - b) Briefly describe the role of statistical methods in geographical studies.
- (3) Kenya Planters Cooperative Union (KPCU) appointed two companies to audit its annual financial accounts for the year 2001. The two companies arrived at different conclusions, and yet they used the same data. In the light of the above statement, comment how statistics might have been misused.

SECTION 2

2.0 Numbers, Measurements and Observations

2.1 Introduction

This section is concerned with two types of numbers, and how these numbers can be rounded to the nearest tens, hundreds, thousands, etc. The distinctions among different types of scales of measurement, which are important in the application of statistical

techniques, are detailed in this section. Finally, the meaning of the concept of observation is also provided.

2.2 Objectives

- a) To provide the meaning and types of numbers
- b) To discuss scale and types of scale of measurements
- c) To define the term observation.

2.3 Numbers

There are two types of numbers: integers and real numbers. An integer is a whole number, with no fraction or decimal part. Integers can be either positive, negative, or zero. Examples of integers are 9, 8, 7, ...0, -5, -4, -3, ...and 0. A real number is any number, which has a fractional or decimal part, and 2.175, 0.6548, -0.0054, 9.99, 111.4532 are real numbers.

Rounding of Numbers

Rounding is an expression of a number or figure into a desired estimated value, for example, to the nearest ten(s), tenth(s), hundredth(s), thousands, and etcetera. A figure is rounded up or raised to the next number of either 10s, or 100s, or 1000 units.

For instance, 125 can be rounded up: $125 = 12 \text{ tens} + \frac{1}{2} \text{ tens}$.

This can be rounded up to 130.

The rounding is adjusted up words downwards depending on the values of the ones, tens, and hundreds in the figure. If the value of ones is 5 or more, the figure will be rounded up wards, and vice-versa, for values less than 5.

Errors can be made when rounding the numbers. This errors can be either biased or unbiased. The biased error (or systematic error, or cumulative error) tends to be all one way, and any figure obtained will be either too small or too big. The latter occurs due to counting to the nearest unit, and thus resulting in figures, which are overstated or understated.

2.4 Measurement

Refers to the assignment of values to particular objects or events. This measurement exists in a variety of forms owing to the diversity of data used by geographers. With

this fact in mind, geographers along with other scientists have found it useful to identify different scales in which data can be measured. These scales reflect a set of all values for observations together with the rules for assigning these values to such observations.

Basically, there are four common scales of measurement, each identifying and using different properties. These scales are;

- a) Nominal,
- b) Ordinal,
- c) Interval,
- d) Ratio.

2.4.0 Nominal Scale

It is sometimes referred to as classificatory scale. Used when placing observations into mutually exclusive categories. It involves categorization with numerical valuation. Its most basic form is the binary, or presence/absence, or a division of a set of objects on a given criterion such as whether or not Nairobi has more than 1 million inhabitants, or a person's sex (male or female).

The nominal scale is the weakest level of measurement, and data based on this scale cannot be manipulated according to the basic arithmetic rules. For instance, if one town is in region 1 and another in region 2, one cannot say that on average they are in region 1.5. Similarly, the sex of human beings cannot be manipulated to have an average of a man and a woman. It has either to be a man or a woman, and thus not otherwise.

2.4.1 Ordinal Scale (Ranking Scale)

On this level of measurement, data can be placed in a ranking position along a continuum. Each individual occupies a separate position on the continuum, or places may be in ordered categories.

This level of measurement is used in the absence of reliable population data. For instance, starting with City of Nairobi, Mombassa, Kisumu...one can rank the ten largest towns in Kenya by size.

The ordinal scale is a stronger level of measurement than nominal scale.

During the ranking of data, two or more observations may have the same magnitude so that one ends up with observations tying for a given rank. Except for tied rankings, the magnitude of the difference between sequentially ordered observations is always 1.

2.4.2 Interval Scale

At this level of measurement, each observation has a separate value for the phenomena being measured. It assigns an exact numerical value, so that the difference any two observations on the scale are precisely known.

Basic arithmetic procedures can be conducted on such data, although some of the results are as not readily interpretable or comparable as are those with data in the next final category. Interval scale lacks true zero points; e.g., 32°F is equivalent 0°C.

2.4.3 Ratio scale

This is the highest level of measurement. All such scales possess a true zero point, as well as permitting precise differences to be calculated, so that measurement retains the same ratio to one another no matter what units are employed. For example, the distances 10 km and 20 km form same ratio (1/2 or 2) as their equivalents, say 16km and 32km. This is not true for say, temperature, which falls under the interval scale.

2.5 Observation

This is information from a unit. It is the value assigned to each item under investigation and is represented by X_i (X subscript i). If the total number of observations is n, then i range from 1 to n. An observation consists of one or more pieces of information about the unit, which had been referred in Lesson one as

variables. The data in Table 2.0 contain some detail on various concepts and definitions regarding observations and variables.

Table 2.0: Household Data

Observation	Marital Status	Age	Type of House	Education Level
1	Married	65	Iron Sheets	Secondary
2	Married	40	Tiles	University
3	Widowed	70	Tiles	University
4	Married	54	Tiles	University
5	Married	42	Tiles	Primary
6	Single	28	Iron Sheets	Secondary
7	Divorced	40	Tiles	University
8	Married	44	Tile	University
9	Married	54	Tiles	Primary
10	Married	35	Tiles	Secondary
11	Married	53	Tiles	Secondary

12	Separated	34	Iron Sheets	University
13	Married	44	Tiles	Secondary
14	Married	54	Tiles	Primary

The data set in Table 2.0 contains observations for households in an estate in the city of Nairobi. In this example, each household is a unit, or observational unit. The set of values of the four variables (marital status, age, type of house and education) is called an observation. Most data sets are arranged as in Table 2, where the observations are given as rows and the variables are represented in the columns.

2.6 Exercise

- (a) Express the following figures to the nearest unit: 175, 123, 837, 438, 100.7, 872.4, 1800.6, 2785.2.
- (b) Round down and up the following numbers: 181, 635, 78, 119, 63.5, 12.9, 15523, 13.5, 318, 126. Determine the biased and unbiased errors in the figures and comment on your results. Using your results, determine the differences between the total rounded values and the totals of unbiased error of the same values.
- (b) In which scale can you measure the following:
- Species of each tree in a forest
 - The religion of students in your class
 - Temperature measured on the Celsius scale
 - Sex of students in your class
 - Concentration of DDT in milligrams per liter.

SECTION 3

3.0 Sources and Methods of Data Acquisition 3.1 Introduction

This section brings to our understanding the term data. We learn the qualities of a data set and types of data to be collected. In addition, census and sampling as procedures of data collection are extensively covered.

3.2 Objectives

- (a) To explain the term data and important aspects related to data.
- (b) To discuss the basic techniques employed in data collection.

3.3 Definition

Data is a plural for the noun datum, meaning a fact (numerical or otherwise) from which a conclusion may be drawn. A datum contains information, whereas a number, an adjective or other form of description may not. The information in a set of data may be contaminated. For example, mixing or confounding with other kinds of information, and thus ceasing to be data.

Statistical data are the raw material of any statistical investigation, and they arise whenever measurements are made or observations are classified (Refund 1979). They may be sizes in acreages of individual farms, or personal characteristics such as age, height, and family size, or they may be simple answers such as “Yes or No”, or descriptions of person’s marital status as single, married, widowed or divorced.

3.4 Aspects of Data Collection

Whenever data are collected, the following questions/aspects require answers:

- a) Why should data be collected and what uses will these data serve? If there is no purpose or no use for these data, why collect them?
- b) What data will be collected? All pertinent data for studying a phenomenon should be collected. Thus it is essential to determine what data to collect in light of why the data are collected.

- c) How and where are data to be collected? The statistician may be helpful in designing and planning the investigation and in determining how and where the data are to be collected. Also involved are how to measure and quantify information on the various phenomena in an investigation. This is especially true of social phenomena.
- d) Who is to collect the data and where are they to be obtained?

3.5 Types of Data Collection

Three main types of data collection are:

- i) Observational investigations or studies,
 - ii) Experimental investigations, and iii)
- Census and Sample surveys.

3.5.1 *Observation*

In observational types of investigation, records are kept on whatever observations are available with little or no thought of making them representative of the population.

3.5.2 *Experiment*

In an experiment, conditions are often introduced which do not appear in the population, and there is a considerable degree of control over the conditions of the experiment. The controlled conditions in an experiment may not be available in observational and survey investigations.

3.5.3(a) *Census*

The term means a “count” usually of people but sometimes of dwellings or of consumer durables, for instance, automobiles, telephones and refrigerators, or wildlife such as elephants and rhinos. In agriculture it may refer to a count of livestock such as cattle, and poultry, or areas under particular types of cultivation.

A human population census is taken in many countries at regular time intervals. In Kenya it is usually taken at an interval of 10 years, and it is during this time that primary source of information about the population of a country is obtained. Although the human population census, considered as a field of operation has much in common with other kinds of social surveys, it nevertheless sharply distinguished by its traditional background, legal sanctions, coverage, and by the whole scale of operation

and the resources normally devoted to it. All these permit a far greater content and depth of analysis than can normally be encompassed in other types of field surveys.

According to official U.N. (1967) a census of population is defined as “the total process of collecting, compiling, evaluating, analyzing and publishing of demographic, economic and social data pertaining to all persons in a country, or a well-delimited part of a country at a specified time.” Certain essential features follow from this definition:

- (a) An official census is sponsored by the government of the area or some larger region within which the area falls;
- (b) The area covered is precisely defined;
- (c) The principle of universality applies, i.e., the enumeration either includes every member of the community to which the census relates without omission, or duplication, or if sampling is used, it must give every member of a stratum equal likelihood of inclusion.
- (d) Relating the census to a point of time implies simultaneous enumeration of all persons included in the census, or at least a close approximation to simultaneity.

Legal basis

There is usually a legal basis for the census in order to render public participation enforceable. The legal sanctions that can be applied distinguish a census from other voluntary types of social surveys, which are likely to suffer from incompleteness and bias.

Publicity

Completeness and accuracy are, however, distinguishing characteristics of the census that depend less on legal sanctions than upon public acceptance of the need for the collection of the information in the interest, ultimately of public welfare. For this

reason, it is necessary to ensure the retention of public confidence by careful prior explanation of the information needs and the subsequent demonstration that all the information that has to be gathered is needed and will be used in the public interest.

Confidentiality and Privacy

Much of the information required in a modern census involves questions bearing upon intra-family and extra-family relationships, and on educational attainment, earning capacity, and employment status. Such questions tend to be regarded as bordering upon an invasion of privacy. To retain public confidence, there has to be a demonstrably valid assurance that the personal records will not be accessible to third parties, and the processing of the records into statistics will not permit the identification of individual persons or households.

Mapping

This means that as part of the census preparation, there is a large-scale and very careful mapping operation in which the entire country is divided into parts (enumeration districts) to be covered by individual enumerators. The delineation of enumeration districts must satisfy the following criteria:

- In the interest of speed of making contacts, and the simultaneity of the whole census operation, no district should be larger than one can, reasonably be covered on foot by an enumerator in 1 or 2 days.
- The boundary of the district must be clearly recognizable on the ground. Each enumerator is given a map of his/her district, and there must be no ambiguity as to where this district ends, and another begins.
- Each district must be completely contiguous with others. There must be no gaps between defined districts, which may lead to under-enumeration, and no overlapping, which may heighten the risk of double enumeration.
- Given that much of the census information is required for local as well as central government purposes, the districts must be designed that they be aggregated to exactly complete local administrative units.

Timing

Population changes are measured by comparison of the census enumeration time with that of another time. It is essential, therefore that the census shows the picture of population size and structure at a fixed point. It is normal to select a time at which population movement is minimal and at which most people are at home within the *de jure* households. In Kenya, the time reference is a round midnight.

Notes: In working with official statistics, the Geographer is often tempted to believing that such data are always totally accurate. However, all sources contain some degree of measurement error. In official sources, quite often the error may take the form of under-enumeration. Sometimes the measurement errors are due to lack of cooperation between organizations or people, and thus resulting in less than a total response to government surveys. In other cases such errors arise owing to a mishandling of data, which is often more difficult to detect.

In any country the government is the main compiler of statistics, which are published in a variety of forms. Despite this recognition of this fact, many government statistics are still not available at either a comparable spatial scale or the same time period. From the Geographer's point of view one of the major problems associated with most published statistics is that the spatial units used for compiling them are rarely meaningful often being irregular and incongruent in shape and size, and in some instances changing over time. There are two possible methods of solving the problems of unsuitable area units: The first of these is to compile the official data in statistically sensible and consistent spatial units. The second method is for the Geographer to collect his/her own data from the field surveys.

Whilst collecting the data one needs to employ methods, which are scientifically proven and less time consuming. Such methods include population census and sample surveys, among many others.

3.5.3(b) Sample Surveys

Sample surveys may be grouped into two broad categories, namely: probability sample surveys and non-probability sample surveys. In the former type, the probability of selecting sampling units from the population is known, whereas it is not in the latter type. Whether or not non-probability sample estimates can be regarded, as a representative of the population parameters is unknown. Despite this, a large number of surveys are of the non-probability type. The lower cost and convenience of such surveys is attractive to investigators even if they are unable to check on the biases in the survey. For geographers they normally rely on the probability types of samples as described below:

1) Simple Random Sampling

This is the most straightforward sort of sample, where elements from a list or other sampling frame are chosen randomly, usually by using random numbers. Each element in the sampling frame is allocated a unique identifying number. Numbers are then read from say a **random number table**, and elements in the frame with those numbers are included in the sample.

Sampling theory assumes that if the same element is picked twice, it will be included twice in the sample. In practice, this will not generally be done since it would reduce the variability in the sample because the two elements included would be identical. The technical term for not allowing the same item to occur twice in the sample is sampling without replacement, once an element has been chosen, it ceases to have another chance of being drawn, and those remaining have a slightly increased chance.

One major demerit of this procedure is that it might happen, since each number has an equal chance of occurrence and therefore all combinations of elements in the frame could theoretically be encountered, then the chosen sample is concentrated in one part of the population.

2) Systematic Sampling

This procedure ensures even coverage throughout a sampling frame. Instead of using random numbers, a regular spacing is employed, taking every k th individual, or the intersection of a regular grid laid over a map. The sampling individual, k is the reciprocal of the desired sampling fraction, i.e. k is N/n . For example, if the sample is to contain 50 elements and the sampling frame lists 5000, one element in 100 would be chosen by setting k at 100. The sample is chosen systematically from a random starting point and this is the only random number, which needs to be drawn. Thereafter, the choice of the components of the sample proceeds simply with every k th one chosen.

Notes: This procedure is not considered strictly random sampling for two major reasons: (a) the method does not enable all possible different samples of a given size to be selected. The selection of members of the sample is not independent because once the first individual has been chosen, the rest are fixed. This violates one of the assumptions on which sampling theory is based; (b) there may be periodicity within the sampling frame.

Stratified Sampling

This is a refined procedure in which the population is divided into sub-sets, and separate samples are drawn from each. This is important if the population is made up of distinct sub-populations of different size or character. For example, if one wants a sample to contain both men and women, it would be sensible to sub-divide the frame on the basis of sex before sampling, rather than run the risk of obtaining a sample, which may be unrepresentative of the population. Also if one expects different areas to have different characteristics, it will be as good to consider them as separate strata, and choose within each stratum.

This technique has several advantages, among them; (a) It tends to improve population estimates, provided that the variable used for classification is related to the subject of the study, and (b) it ensures that each part of the population is represented with a sample of sufficient size for the comparison to be made.

Notes: There is no need for the strata to be of identical size and there is no rule about the appropriate number of strata, although Kish (1965 pp.102) suggests that between 2 and 10 will normally be the best. The researcher should also know his/her study area quite well before commencing the formal data collection stage.

3) Cluster Sampling

The procedure is normally useful when the population is widely dispersed, and therefore reducing a great of travel. It saves a lot of time to sample from carefully chosen clusters. This will be particularly applicable for opinion polls in order to represent different tribes in Kenya, which also include major urban areas, suburban areas and rural areas of varying location, social structure, etc. A multi-stage sampling procedure is thus used involving (a) first choosing representative clusters, which sometimes may be subjectively selected, and (b) selecting a random, systematic or stratified sample within each cluster.

Notes: *There are certain 'ground rules' to be observed when sampling; the most important being:*

- (a) Sampling frame should be available which in some way lists all the entities (e.g. persons, places, and regions) from which the sample is to be drawn, and*
- (b) Measurements should be obtained with respect to entities, which have been selected by the procedure of probability random sampling, or by some satisfactory approximation to it.*
- (c) Although sampling procedures are designed to minimize subjective choices, the selection of sampling frame and sampling method does require common sense. There is no set of rules that will cover every sampling problem, and an investigator must be prepared to use his ingenuity in devising procedures that*

will provide reasonably unbiased data from a reasonable expenditure of money, time and other resources.

Advantages and Disadvantages of Sampling

Advantages:

- (a) If the population of measurement is infinite, we have no other choice, and therefore a considerable amount of time, money and other resources will be saved.
- (b) The time, money and other resources spared will be devoted to ensuring the quality of data, perhaps by training interviewers and fieldworkers properly, or employing professional agencies, and instituting careful checks to reduce the number of errors liable to occur at various stages in the process from initial collection to final tabulation or statistical analysis.
- (c) Results will also be readily available more rapidly from a survey of modest size, an important consideration to planning departments.
- (d) Sampling is desirable if the very act of obtaining measurement is reactive or destructive
- (e) If all surveys are based on the entire populations, individuals will become uncooperative because of being constantly being pestered.
- (f) Some measurements of variable characteristics require restricted sample sites.

Disadvantages:

Sampling error represents the difference between a sample estimate and a population parameter. Unless precautions are taken, sample estimates may be biased. To ensure minimum bias and maximum precision given the resources at the disposal, one must use a procedure based on independent random sampling or probability sampling. If such procedures are used, the probable size of sampling error associated with an estimate can be determined.

3.6 Key Words

Sampling Frame is a list of all entities from which a sample is selected.

Sampling Method is a method of selection of entities from a sampling frame.

3.7 Exercise

- (a) Using examples show why geographers make use of samples.
- (b) Define the following terms as concisely as possible: (i) Stratified sampling, (ii) Cluster sampling, (iii) Periodicity in a sampling frame.
- (c) What features make Census operation different from other techniques of data collection?

SECTION 4

4.0 Organizing Data 4.1 Introduction

When data are collected, the information obtained from each member of a population or sample is recorded in a sequence in which it becomes available. This sequence of data recording is random and unranked. Such data before they are grouped or ranked are called raw data. In order to organize and display such data tables and charts are constructed, which provide effective visual demonstration of such data.

4.2 Objectives

- (a) *To learn how to classify data by type, organize data into tables and summarize data with graphical displays.*
- (b) *To examine ways of analyzing and interpreting the data.*

4.3 Introduction.

The collection of statistical data has grown at such a rate that it requires that the information be disseminated in a summarized form. The development of electronic computers has made it possible to put large masses of data into a usable form in minutes what previously would have taken months or even years. The most common method of summarizing data is to present them in frequency distributions, tables and charts.

(a) Frequency Distributions

When one is dealing with large sets of data, grouping the data into a number of classes can often convey a good overall picture and sufficient information.

Table 4.1 provides survey results of water availability for livestock in 110 farms in Kisii District.

Table 4.1: Distance From Watering Points and Number of Farms in Each Class

<i>Distance in Kilometers</i>	<i>Number of Farms</i>
0 - 2.0	48
2.1 - 4.0	34
4.1 - 6.0	11
More than 6.0	17

Source: Adopted from Osebe (1990): Ecological and Socio-economic Factors Influencing Livestock Production ... (in) Kisii District, West Kenya, pp.69.

Table 4.1 is called a frequency distribution, and it shows how the distance from watering points and the number of farms is distributed among the chosen classes. Since the data are grouped according to their numerical size, one can also refer to such a table as a **quantitative distribution**. In contrast, tables such as the one in Table 4.2, in which data are grouped into non-numerical classes, are called **qualitative distributions**. In general, data measured on nominal scale is referred to as qualitative data, whereas one based on metric scale (ratio and interval) is called quantitative data.

Table 4.2: Estimated Value Per Hectare for selected commodities in Kenya

<i>Commodity</i>	<i>Value Per Ha. (Kf/ha.)</i>	<i>Rank</i>
Coffee	1489	1
Tea	1325	2
Vegetables	913	3
Pyrethrum	419	4
Fruits	295	5
Root crops	205	6
Maize and beans	153	7
Milk	70	8
Millet and sorghum	48	9

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Source: Republic of Kenya (1986): Sessional Paper Number 1.

MWALIMU GREG.

Although frequency distributions present data in a relatively compact form, give a good overall picture, and contain information, which is adequate for many purposes, there are evidently some things, which can be obtained from the original raw data that cannot be obtained from a distribution. With reference to example 1, for instance one cannot determine the exact size of the shortest and longest distance in any of the categories. Similarly, in Example 2, one cannot tell whether the returns from the farms are for one season or one year.

Nevertheless, frequency distributions present raw data in a more usable form, and thus the loss of certain details is usually a fair exchange (Freund 1976).

Steps to follow in the construction of a numerical or quantitative distribution:

- (i) Choose the classes into which the data are to be grouped,
- (ii) Sort out (tally) the data into the appropriate classes,
- (iii) Count the number of items in each class, and (iv)

Display the results in a chart or a table.

The choice of the number of classes and deciding from where to where each class should go requires some rules that are generally considered „sound practice.“ (a)

The choice of the number of classes should depend on the number of measurements or observations that one has to group.

- (b) Each item should go into one and only one class. To this end one must ensure that the smallest and largest values fall within the classification, none of the values fall into possible gaps between successive classes, and the classes do not overlap.

- (c) Whenever possible, make the classes cover equal ranges of values.

Note that in Table 4.1, the last class of more than 6.0 is open. The open class labeled “more than...” serves to simplify the overall picture presented by a distribution when a few of the values are much greater, (or much smaller) than the rest. **b) Graphs and**

Charts

Graphs of frequency distribution are used to clarify and emphasize trends that are not readily discernible in tables. They enable one to estimate some values at a glance and provide one with a pictorial check on the accuracy of the solutions. When frequency distributions are constructed primarily to condense large sets of data into an easy to understand form, it is advisable to present them graphically, i.e. in a form, which has a visual appeal.

The most common kind of graphical presentation of a frequency distribution is the **histogram**. Histogram is a Greek word meaning a cell. A histogram is a series of rectangles, each proportional in width to the range of values within a class and proportional in height to the number of items falling in the class. They are constructed by representing the quantities which are grouped on a horizontal scale, the class frequency on a vertical scale, and drawing rectangles whose bases equal the class interval and whose heights are determined by the respective class frequencies as shown in figure 4.0.

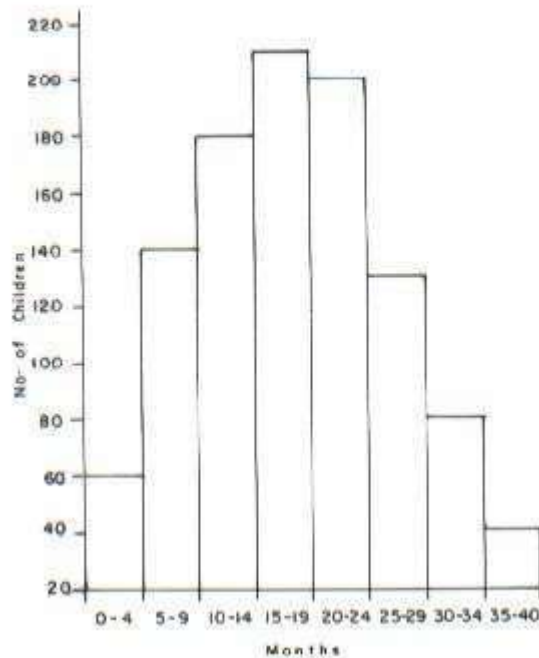
To construct a histogram, the following steps are undertaken:

Step 1: Choose a basic class interval of a given grouped data.

Step 2: Compute the height of each rectangle (bar), i.e., $h = \text{frequency} / \text{class interval}$.

This is because the frequency is proportional to the area of each bar.

Fig. 4.0: Histogram showing true age limits and number of children falling in each age category



It is important that all classes be of the same size, and the class frequencies be represented by the areas of the rectangles instead of the heights. The practice of representing class frequencies by means of areas is essential if one wants to approximate histograms with smooth curves. The practice of representing class frequencies by means of areas is essential if one wants to approximate histograms with smooth curves. Histogram is an important statistical technique used to display some data. It is, for example, from the histogram that a normal curve can be derived.

Similar to histograms are **bar charts** as the one shown in figures 4.1 and 4.2. The class frequency is either vertically or horizontally laid, and the quantities (intervals) on the vertical or horizontal axis, respectively. Sometimes, bar charts are figures whose lengths of the bars are proportional to the class frequencies, and they do not pretend of having a continuous horizontal scale.

Fig 4.1: Bar Chart of Farmers' Age Groups, Where Frequency and Class Interval are on Vertical and
horizontal axis, respectively

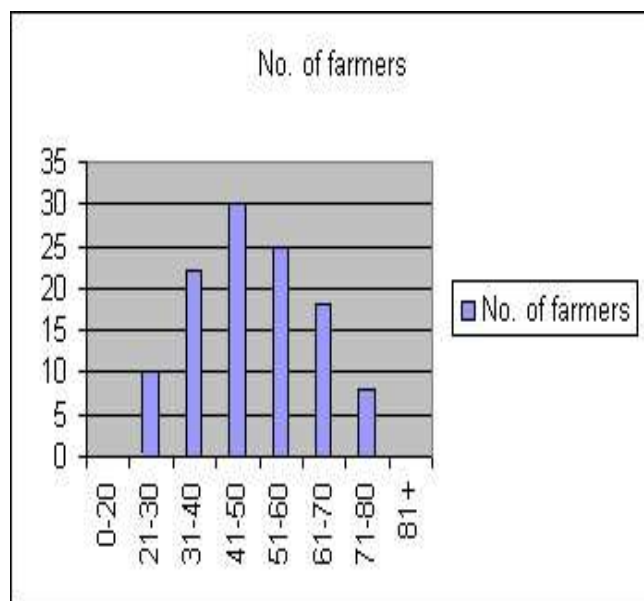
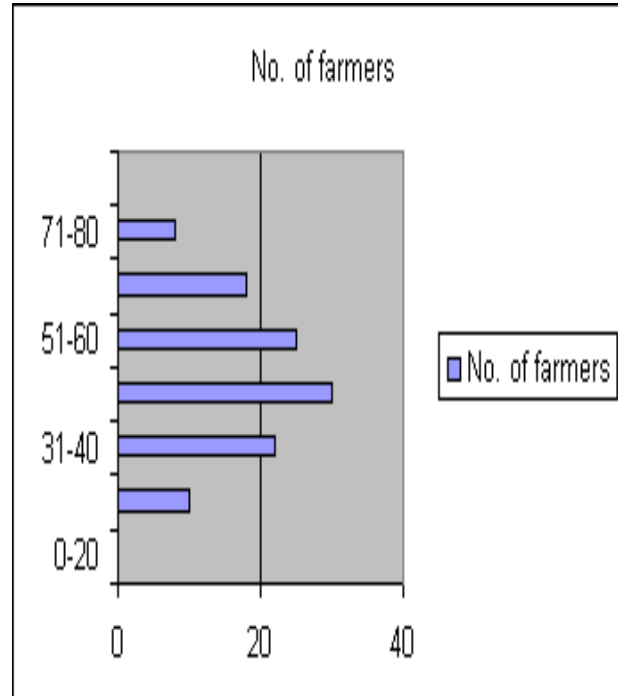


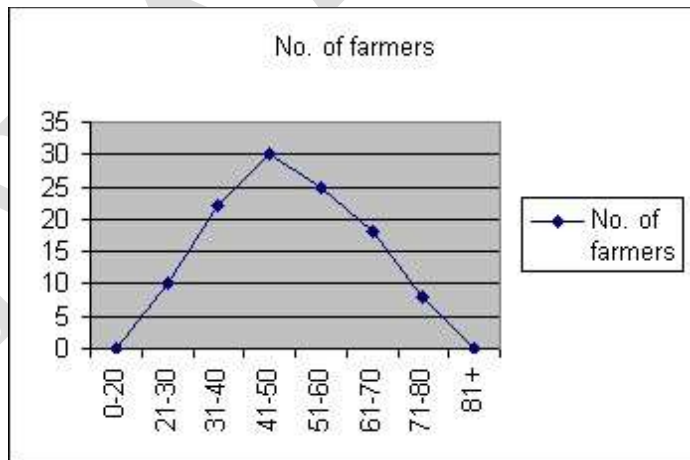
Fig 4.2: Bar Chart of Farmers' Age Groups, Where Frequency is on the Horizontal

Axis



Another kind of graphical presentation of a frequency distribution is the *frequency polygon*, an example of which is shown in Figure 4.3.

Figure 4.3: Frequency Polygon of Farmers' Age Groups



Here the class frequencies are plotted at the class marks, and the successive points are joined by means of straight lines. Note that a class with a zero frequency is added at each end of the distribution to „tie down“ the graph to horizontal scale.

Histograms and frequency polygons are similar. The advantages of histograms are as follows:

- a) The rectangle clearly shows each separate class in the distribution.
- b) The area of each rectangle, relative to all other rectangles shows the proportion of the total number of observations that occur in that class.

The frequency polygon, however, have certain advantages too. They are:

- a) The frequency polygon is simpler than its histogram counterparts.
- b) It sketches an outline of the data pattern more clearly
- c) It becomes increasingly smooth and curve-like as one increases the number of classes and the number of observations, thus taking a shape of a normal curve distribution.

Ogive. This is a graph of cumulative frequency distribution, which enables one to see how many observations lie above or below certain values, rather than merely recording the number of items within intervals. Using the data in Table 4.3, an Ogive is drawn (Figure 4.4). However, it should be observed that in an Ogive, the cumulative frequencies are not plotted at the class marks.

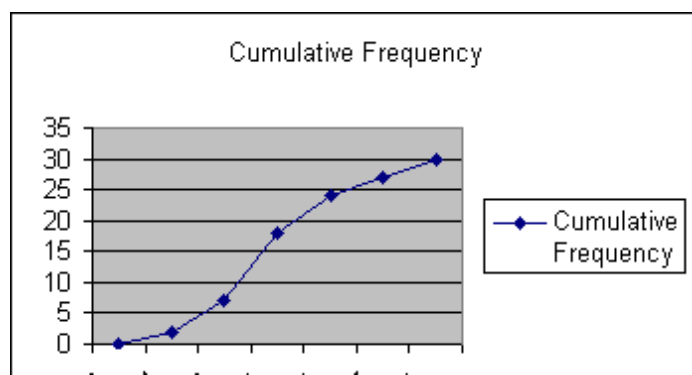


Table 4.3 Cumulative Frequency Distribution of weights in grams

Class	Cumulative Frequency
<15.2	0
<15.5	2
<15.8	7
<16.1	18
<16.4	24
<16.7	27
<17.1	30

Fig 4.4 An Ogive derived from Table 4.3 (above) of weights (grams)

4.4 Key Words and Concepts

Class limits. These are the smallest and largest values that can go into any given class. The class limits can be divided into lower and upper class limits. In Example 4.1, the lower class limits are 0, 2.1, 4.1 and >6.0, and the upper class limits are 2.0, 4.0 and 6.0.

4.5 Exercise

- (1) Construct histograms for the following data and superimpose frequency polygons on them showing the number of motor vehicles passing through toll stations, A and B in one week.

No. of Motor Vehicles	Toll station A	Toll Station B
50-59	15	43
60-69	25	99
70-79	40	54
80-89	108	40
90-99	92	14
100+	20	0

(a). By means of Ogives compare the distributions of motor vehicles as recorded at the two stations.

(b). From the Ogives determine:

- (i) The frequency distribution below 70 and 90 vehicles for toll stations, A and B, respectively, (ii) The class limits below 50%.

(2) Given grouped data below:

x	0-1	2-7	8-15	16-39	40-75	76-100
y	5	10	14	48	8	2

Display the above data using histograms and comment on your results.

(3) According to National Population Census of Kenya for 1999, the Kenyan population in terms of age and sex is recorded as follows:

Age	Male	Female	Total
0-4	1911216	1888827	3800043
5-9	1744366	1725292	3468938
10-14	1504044	1485648	2989692
15-19	1177989	1704712	2378696
20-24	989594	1013340	1902934
25-29	782474	847287	1629761
30-34	583173	575651	1159434

35-39	464956	457942	918892
40-44	367934	304244	233178
45-49	235906	293405	574533
50-54	179017	240657	476523
55-59	150496	180055	360172
60-64	113690	167901	318397
65-69	82966	116980	230670
70-74	66600	91212	174175
75+	82210	60476	176280

Draw the histograms to show:

- (a) Age-male distribution
- (b) Age-female distribution.

Compare your results in (a) and (b) above and comment on them.

SECTION 5

5.0 Measures of Central Tendency

5.1 Introduction

This section provides concise, and consistent descriptions of sets of data employing least function of statistics. Some of the measures such as „average“, or „typical“ characteristics are in common everyday use. Measures of central tendency (mode, median and arithmetic mean) tell us what we can expect a typical or middle data point to be.

5.2 Objective

- (a) To describe the most common measures that indicate where the center or most typical value of a data set lies.

5.3 Introduction

The term Central tendency and average are used synonymously. They provide a single number to describe the general magnitude of all observations in a data set. The most common types of statistical averages are the mode, median and the arithmetic mean.

Whilst these statistics have some properties in common, they each give a different measure of Central tendency, and are used in varying circumstances.

5.3.1 The Mode

This is the any value occurring most frequently in a set of observations. In terms of frequency distribution, the mode is the value at which the graph reaches its maximum height.

A mode or a modal class may or may not exist as follows:

- (i) When there is a single dominant frequency, the distribution can be described as ***uni-modal***,
- (ii) When there are two equally dominant frequencies, the distribution is called ***bi-modal***,
- (iii) In cases where several modes or modal classes occur, then one might be inclined to describe such a description as ***multi-modal***. However, one comes to a position where it is more realistic to say that no mode at all exists.

Table5.1: Age groups of pupils in a Primary School

10, 09, 08, 12, 12, 07, 12, 12, 06, 05, 11, 12, 14, 12, 04, 15, 07, 12, 10, 12.

In table 5.1 above, the mode is 12, which is the value appearing eight times, more than any other value in the table.

Table 5.2: Figures of Minimum Temperature for the first fifteen days of August 2000

7.3, 10.7, 9.1, 8.4, 13.9, 9.4, 6.1, 10.8, 8.9, 8.8, 10.6, 11.2, 7.7, 9.7, 9.9.

In table 5.2 above, no value occurs more than once, and thus one can conclude that no modal class can be identified. More commonly with this kind of data, a mode can be obtained by arranging the observations into classes. Doing this and ensuring the class intervals are of equal size, one can obtain:

Table 5.3 Class interval of Minimum Temperature for Fifteen Days of August 2000

Class Interval	Observations	Frequency
6.0 - 7.9oC	6.1, 7.3, 7.7	3
8.0 - 9.9oC	8.4, 8.8, 8.9, 9.1, 9.4, 9.7, 9.9	7
10.0 - 11.9oC	10.6, 10.7, 10.8, 11.2	4
12.0 - 13.9oC	13.9	1

From these data (Table5.3) the modal class, 8.0-9.9oC can be identified and if the grouped data are plotted as a histogram, the modal class stands out as the peak in the distribution at a frequency of 7.

The mode is useful in describing data measured on the nominal scale. It is also an appropriate measure of central tendency in situations where one is catering primarily to the needs of the most populous category in a distribution. However, the dataset may have no single frequently occurring value, for example, in bi-modal and multi-modal distributions.

5.3.2 Median

The median is the middle item/observation in a set of data arranged in ascending or descending order. In other words, it is the point, which separates the upper 50% from the lower 50% of the observations. It is represented by one of the actual values in an odd-numbered distribution. In a case of an even-numbered distribution, adding the values of the two middle observations and divide the sum by two derives the median.

The median is appropriate measure of central tendency for ordinal scale data.

5.3.3 The Arithmetic Mean

To derive the mean, all observed values of x_i are summed and then divided by the number of observations (n) as shown in the following equation.

$$\bar{X} = \sum x_i / n$$

Where $\bar{X}$ = mean

$\sum x_i$ = sum of all values of x_i

n = number of observations.

The mean reflects the magnitude of every individual value and any data set can only have one mean. Unlike the median, the observations need not be ordered. For this reason, the mean is an important and commonly used measure of central tendency.

The significance of the mean goes beyond its immediate measure of central tendency, since it possesses a number of significant properties such as:

- a) the sum of deviations from the mean is always zero. In other words, the amount by which high values exceed the mean are equaled by those which fall below,
- b) the sum of the squares of the deviations from the mean is at a minimum.

5.4 Exercise

- (a) Explain in detail the purpose of a measure of central tendency.
- (b) Name and describe the three most important measures of central tendency.
- (c) Determine the mean, median, and mode for the data below:

37, 28, 36, 33, 37, 43, 41, 28, 24, 44, 27, 24, 37, 40.

SECTION 6

6.0 Measures of Dispersion

6.1 Introduction.

Although measures of central tendency (location) describe one important aspect of a set of data- their „middle“ or their „average“ and, are generally considered to be the most important characteristics of a distribution, measures of dispersion of the values are also very important. The procurement of any characteristic of measures of dispersion is based on the premise that values in a set of data are not all alike. The extent to which they are unlike or vary among themselves is thus of basic significance in statistics. Consequently, the need for other ways of measuring the extent to which data are dispersed, or spread out, and the statistical measures which provide this information are called *measures of variation*.

6.2 Objective

- (a) To describe the different types, that measure the amount of variation or spread in a data set.

6.3 Types of Dispersion

Some of the most frequently used measures of dispersion are range and standard deviation.

- (a) **Range.** This is the simplest and intuitively most obvious measure of dispersion. It is defined as the difference between the largest and smallest observed values.

Although conceptually simple, the range has one serious weakness; ignoring any information from all the other values in a data set.

- (b) **Standard deviation.** By far, this is generally the most useful measure of variation. Its computation is based on the set of distances or differences of the observed values of say y from the mean, $\bar{y}$.

These differences, $(y_i - \bar{y})$, $i = 1, 2, \dots, n$, are called the deviations from the mean. It should be noted that the calculation of the standard deviation, always involve the use of the arithmetic mean, and is a measure of the dispersal of the values around the mean.

The formula for its calculation is as follows:

$$s = \sqrt{\frac{\sum (y - \bar{y})^2}{n}}$$

Where s = standard deviation

n = no. of observations

Sigma sign (Σ) = advises one to add the squares of differences between each value of y and $\bar{y}$ (mean).

Steps: (1) Set out the data in a column, (2) Calculate the sum (Σy) and the mean of y to get $\bar{y}$, (3) Subtract $\bar{y}$ from each of the observations in turn forming another column of values, which are deviations from the mean, (4) Square the deviations, forming a third column, (5) Add the squared deviations together forming $\sum (y - \bar{y})^2$, (6) Divide $\sum ((y - \bar{y})^2)$ by n . This gives a value known as the variance. (7) Take the square root of the variance to get the standard deviation, s Table 6.0).

We can use the following hypothetical data to compute the standard deviations:

Table 6.0 Hypothetical Data

--

X: 6, 7, 4, 5, 1, 5, 6, 8, 6, 2. Y: 5, 4, 6, 4, 2, 3, 4, 0, 7, 8.

Example 6.1

X (Step 1)	(Step 3) $X - \bar{X}$	(Step 4) $(X - \bar{X})^2$
6	1	1

7	2	4
4	-1	1
5	0	0
1	-4	16
5	0	0
6	1	1
8	3	9
6	1	1
2	-3	9
$\Sigma X = 50$		
(Step 2)		(Step 5)
$\Sigma X = 5$		$\Sigma (X - \Sigma X)^2 = 42$
Var. = $\Sigma ((X - \Sigma X)^2) / X = 42/5 = 8.4$ (Step 6) Std deviation = Square root of the variance. Square root of 8.4 is 2.9 (Step 7).		

Example 2

Y (Step 1)	Y - ΣY (Step 3)	(Y - ΣY) ² Step 4
5	1	1
4	0	0
6	2	4
4	0	0

2	-2	4
3	-1	1
4	0	0
0	-4	16
7	2	4
5	1	1
$\Sigma Y = 40$ (Step 2) $\Sigma X = 4.0$		$\Sigma (Y - \bar{Y})^2 = 31$ (Step 5)
Var. = $\Sigma ((Y_i - \bar{Y})^2) / \Sigma Y$ $31/4 = 7.75$ (Step 6) std deviation = square root of the variance, Square root of 7.75 = 2.78 (Step 7).		

Notes:

Large magnitudes of deviation imply a high degree of variability, and small magnitudes of deviation imply a low degree of variability.

If all deviations are zero, the data set exhibits no variability; that is all values are identical.

The mean of these deviations would seem to provide a reasonable measure of dispersion. However, a relatively simple exercise in algebra shows that the sum of these deviations, that is, $\Sigma (Y - \bar{Y})$ or $\Sigma (X - \bar{X})$ will be zero. Thus this quantity is not useful.

One way to neutralize the effect of opposite signs is to base the measure on variability of the squared deviations. This criterion is the basis for the most frequently used measure of dispersion, the variance.

(c) **Coefficient of Variation (Cv)**

It measures the variability of a dataset (observations). It is expressed as a ratio of the population standard deviation, σ and the sample mean, $\bar{x}$.

Thus $Cv = 100 \cdot \sigma / \bar{x}$, tells us the dispersion of data compared with the sample mean as multiplied by 100. The variability is expressed as a percentage.

Example

Calculate the coefficient of variation given $\sigma = 1.5$ and $\bar{x} = 20$.

$$\begin{aligned} Cv &= (100 \cdot 1.5 / 20)\% \\ &= 7.5\% \end{aligned}$$

This means that the dispersion of data with respect to the mean is only 7.5%. Note that when $\sigma = \bar{x}$, $Cv = 100\%$, i.e., there is 100% variability in the data set with respect to the mean sample, or the data vary greatly with respect to the mean of the sample. **(d)**

Chebyshev's Theorem

Although the mean and the standard deviation can be used to describe any distribution, no matter what the shape may be, a much more conservative relation between the distribution and its mean and standard deviation is given by Chebyshev's theorem.

The theorem states thus: For any set of data (population or sample) and any constant k greater than 1, at least $1 - 1/k^2$ of the data must lie within k standard deviations on either side of the mean.

Example

If two samples have the same mean, $\bar{X} = 140$, what can one say about the percentages of the values that must lie between 125 and 155, given that their respective standard deviations are $s=10$ and $s=1.5$?

For the first population, the values must lie within $15/10=1.5$ standard deviations of the mean, and for the second population the values must lie within $15/1.5=10$ standard deviations of the mean.

Thus, at least $1-1/(1.5)^2=5/9$ or about 55.6%, of the values of the first sample must lie between 125 and 155.

While the corresponding percentage of the second sample is at least $1-1/(10)^2=99/100$, or 99 percent. This demonstrates how the magnitude of the standard deviation “controls” the concentration of a set of data about its mean.

6.4 Key Concepts

The standard deviation of a set of observed values is defined to be the positive square root of the variance. This measure denoted by s and does have a very useful interpretation as a measure of dispersion. For the two example data sets, the standard deviations are:

Data set X: $s = 2.90$

Data set Y: $s = 2.78$

Although the mean and standard deviation are only two descriptive measures, together the two provide a great deal of information about the distribution of an observed set of values. This is illustrated by the **empirical rule**. This rule is stated: If the shape of the distribution is nearly bell-shaped, the following statements hold:

The interval $(\bar{y} \pm s)$ contains approximately 68% of the observations.

The interval $(\bar{y} \pm 2s)$ contains approximately 95% of the observations.

The interval $(\bar{y} \pm 3s)$ contains virtually all of the observations.

In each of the intervals, the mean $(\bar{y})$ is used to describe the central tendency, and the standard deviation (s) is used to describe the dispersion of a given portion of the data.

6.5 Exercise

1. An airline's records show that its flights between two cities arrive on the average 4.6 minutes late with a standard deviation of 1.4 minutes. At least what percentage of its flights between these two cities arrive anywhere between: (a) 1.8 minutes and 7.4 minutes late? (b) 1.0 minutes early and 10.2 minutes late? (c) 3.8 minutes early and 13.0 minutes late?

2. A study of the nutritional value of a certain kind of bread shows that on the average one slice contains 0.260 milligram of thiamine (vitamin B1) with a standard deviation of 0.005 milligram. Between what values must be the thiamine content of

a) At least 35/36 of all slices of this bread?

b) At least 63/64 of all slices of this bread?

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7.0 Introduction to Cartography

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7.1 Introduction

The section begins with the definition of term cartography. It explores nature of the discipline, which has been shown to have very interesting dimensions. It also provides a general awareness of the changes and developments during the long history of mapmaking. Thus giving the reader not only an understanding of the roots of the profession, but also providing a background that serves to emphasize how profound are the developments of the present cartography.

7.2 Objectives

- a) To define cartography, and
- b) To provide a historical background of cartography.

7.3 Introduction

Cartography has been defined as the art, science and technology of making maps, together with all stages of evaluation, compilation, design and drafting to produce a new or revised map document from all forms of basic data. It also comprises all the stages in the reproduction of maps. From this definition it is clear that:

- (a) Cartography is concerned with the design and presentation of data (but not collection). This distinguishes it from surveying and photogrammetry.
- (b) It is a series of techniques, rather than a consistent whole.
- (c) It involves careful design of instruments for recording, calculating, displaying, analyzing, and in general understanding the interrelationships of objects in their spatial context.
- (d) Bring objects into view.

According to *The Multilingual Dictionary of Technical terms in Cartography*, the word Cartography is defined as “the art, science and technology of making maps, together with their study as scientific documents and works of art. In this context, maps are regarded as including all types of maps, plans, charts and sections, three dimensional models of globes representing the Earth or any celestial body at any scale.” In broad terms, Cartography includes any activity in which the preparation and use of maps is a matter of basic interest. This includes (i) teaching the skills of map use (ii) Studying the history of cartography (iii) maintaining map collections and associated cataloging and bibliographic activities, (iv) designing and constructing maps, charts, plans and atlases.

The British Cartographic Society has defined cartography as a study concerned with all stages of compilation, design, draftsman and evaluation required to produce a new

map or revised documents of maps. It encompasses the study of maps, their historical evolution, methods of cartographic presentation and use of maps. The British concept of cartography agrees with other above definitions, which exclude land and hydrologic surveying and photogrammetry, although these subjects are closely linked, and considered as being inseparable.

7.4 Significance of Maps

Cartography consists of a group of techniques fundamentally concerned with reducing the spatial characteristics of a large area or a portion or all of the earth, or another celestial body, putting it in a map form to make observable. It is this map that has the objective of serving as a means of communicating spatial relationships and forms.

7.5 Nature of Cartography

Cartography is a science, an art, and technology.

It is a science because:

- It employs scientific principles in the form of reason and logic in constructing its products. The principles are derived through the analysis of scientific data. It has its foundations in the sciences of geodesy, geography and psychology.
- It contains visual relationships inherent in this form of expression. With the applications of the principles and laws governing these reactions, the cartographer designs his products to fit these perceptual patterns.

Cartography can be described as an art and technology because of the following:

- It emphasizes efficiency with which the communicative objective is obtained.
- It is concerned with the production of beautiful maps (map design aesthetic), with a lot of artistic appeal; in this respect, making the discipline definitely an art and technology.
- It is also a creative art in the way that careful, creative literary expression is used in art.
- Cartography is considered as a technology for producing maps. Those who take a technological view point to many concrete advances in the field. Manual

skills that have been largely replaced by optical-mechanical technologies, and which in turn have given way to photochemical techniques. Currently, electronic methods are revolutionizing the field.

7.6 The History of Mapmaking

The history of mapmaking can be divided into four historical phases, namely Early, Medieval, Renaissance, and Modern Age phases.

(a) *Early Phase*

No one knows when the first map was made, but it is thought to be at least as old as communication of written language. One of the oldest authentic maps to survive is a clay tablet made nearly 5000 years ago, depicting mountains, water bodies and other geographic features in Mesopotamia. Topographic maps, which depict detailed features of a limited area, seem to be the earliest maps made in all societies. They would have been useful and practical in the era before there was widespread travel and before people began to be curious about the Earth as a whole (Robinson et.al 1984). Only a few fragments of such maps have survived from ancient periods in eastern and Western cultures.

According to W.A. Rickett and Kuan-Tzu (1965) topographic maps matured earlier and attained a higher level of appreciation in the Eastern than in the Western world. In China, maps made from silk were made as early as the Second B.C. These topographic maps show some streams, roads, mountain ranges, and identified over one hundred items by name. These maps were sophisticated in that the concept of scale, symbolism and orthogonal viewpoint were well developed.

One distinct feature of early Chinese cartography was the use of a square grid over the map area. In the grid, lines were not latitudes and longitudes, but just lines made up of equally spaced, orthogonal straight lines. Chang Heng is credited for the use of square grid in the Second Century. Pei Hsiu (A.D. 224-271) is also considered to be the father

of Chinese cartography who enunciated six principles to be followed in mapmaking. The first three are that the scale should be defined by using graduated divisions, the square grid to be used to depict correct locations and that right-angled triangles be used to derive distances. The last three deal with problems of determining positions in uneven terrain and locating them properly on the plane surface of the map.

In this early phase of cartography, China was technologically more advanced than the Western world. The compass was used much earlier, and paper, which was invented in the Second Century A.D. in China, was to be adopted in the West after 1000 years.

According to Robinson et.al (1984) the system of describing positions on the earth by latitude and longitude and the division of the circle into 360° was well established by the Second century B.C. Estimates of the size of the earth were made by the ancient scholars Eratosthenes (ca. 276-195 B.C.) and Posidonius (ca. 130-50 B.C.) from angular observations of the sun and stars in the Eastern Mediterranean area. The methods used were entirely correct, but the required assumptions and precision of their observations were wrong.

Writings by Claudius Ptolemy (ca. A.D. 90-160 A.D.) apparently reflect clearly a record of advanced stage to which cartography had developed by the end of the Greek period. Ptolemy lived and worked in Alexandria, Egypt, which had become an intellectual center of the Western world. He brought together into a book simply referred to as *Geography*. It contained what was apparently then generally known concerning the earth. The *Geography* included among other things a treatise on cartography. Ptolemy's writings were lost to the European world for more than 1000 years, but fortunately they were preserved by the Arabs and later came to light again in Europe. From the descriptions the Ptolemaic maps were reconstructed and they had a profound influence on European geographic and cartographic thinking during the Renaissance. Marinus, a contemporary of Ptolemy who was a cartographer-geographer also wrote and made maps.

(b) The Medieval Phase

The earlier part of the medieval period in the Western world is called the Dark Ages, and it was not until after the first millennium of the Christian era that enlightenment began to show in the field of cartography and geography. Three advances are noted (1) the contributions of Arabs, (2) the growth of interest in faraway places, and (3) the remarkable development of portolan charts.

The Arabs had preserved the writings of Ptolemy, translated them, and made several attempts to measure the size of the earth in the 9th century A.D. The culmination of Arab cartography was reached in the work of Idrisi (A.D. 1099-1164), an Arab scholar at the court of King Roger of Sicily. His world maps made in A.D. 1154 and 1161 were a great advance over contemporary non-Arab works.

One of the more remarkable occurrences in the history of mapmaking is the sudden appearance in the 13th century of quite accurate sailing charts called portolan charts. They were the products of the experience of a large amount of open water and coastwise sailing in the Mediterranean and adjacent areas. Characteristically, they were covered by a systematic series of unlabeled, intersecting, straight, directional lines radiating from wind, and later compass roses. The lines and shapes of the bounding coasts of seas on these charts were surprisingly accurate but this accuracy did not seem to extend to maps of the land behind the coasts until much later. Although many portolan charts must have been made, none earlier than the late 13th century has survived. This phase in cartography came to an end during the 15th century.

Renaissance Phase

The Age of discovery, centering on the monumental achievements of Columbus, da Gama, Cabot, Magellan, Elcano and others at the end of 15th century rekindled interest in the mapmaking and publishing, which soon became a lucrative calling. By the latter half of the 16th century, the profession was generally in good standing and was well

supported, even though its products were still far from being first-class examples of objective scientific thought.

One of the circumstances that contributed greatly to the rapid advance of cartography was the invention in Europe shortly after 1450 of printing and engraving, which soon made possible the reproduction of maps in numerous copies. Great map publishing houses such as Mercator, Blaeu, Hondius and others in France and Holland rose and flourished. Except for religious and some navigational data, the mapping of any geographic information beyond what is referred to today's base data was unknown. The mapping of most other kinds of geographical distributions had to wait for the inquiring minds that wanted such maps.

Between 1600 and 1650 a new and fresh attitude toward cartography arose. For the first time, accuracy and scientific method became fashionable again. In the second half of the 17th century, the French Academy of Sciences was founded with one of its avowed purposes being the improvement of charts and navigation, which included cartography among its important concerns. This was the period when precise navigation in the open oceans had become a serious problem, and thus necessitating the solution on accurate determination of the size and shape of the earth and on the development of a practical method for determining the longitude. In the middle of the 18th century, the French initiated a detailed topographic survey of their country at a scale of 1:86,400 (about 1.25 miles to an inch) and almost completed it prior to the end of the century. Harrison's chronometer for longitude determination was perfected in England in 1791, and the relatively rapid production of maps followed.

The problem of representing land-surface form more precisely arose, and soon devices such as the hachure and contour were developed. Topographic maps had covered a large proportion of Europe by the last half of the 19th century. Although these maps were neither cheap, nor having a wide distribution, they formed a foundation on which the cartography of the land that followed was to be based. Another development was the official production of charts, whose production had been largely in the private

hands. In the 19th century great modern governmental hydrographic offices were established to produce the detailed charts needed for safe use of oceans in Spain and Portugal. These counterparts to the great land surveys produced navigational charts, bathymetric maps and a variety of special purpose maps.

Two major developments occurred in the Renaissance phase. The metric system of measurement, which provided a kind of a natural universal unit to which every other unit could be compared, was introduced. Soon thereafter, in the early 19th century, map scales were stated in fractions or proportions in which one unit on the map represented so many of the same units on the earth, e.g., 1:100,000. The use of metric system led to round number scales such as 1:25,000 instead of odd scales such as 1:63,360 (1 inch on the map represents 1 mile on the earth).

Beginning slowly in the latter part of the 17th century, thematic maps were added to the repertoire of cartography. Thematic cartography had sketchy beginnings and had to wait for the broadening of scientific inquiry and the study of people and their institutions, which grew rapidly especially in the 19th century. This required a lot of mapping. The variety of subject matter mapped after thematic cartography developed was very great. Of particular interest in the history of thematic mapping is Edmund Halley's publication in 1701 of a map showing the distribution, as then known of the variation of the compass, i.e., the angular difference between true north-south and the alignment of the compass needle. Most important to the development of thematic cartography was the branching out of science into a number of separate fields such as the physical geographer, geologist, meteorologist, and biologist, and the investigators now called social scientists. The rapidly growing earth, life and behavioral sciences needed maps, and in general they required the smaller-scale compiled thematic maps.

Modern phase

Cartography has probably advanced more technically since the beginning of the 20th century than during any other period of comparable length. Many factors combined to

promote this phenomenal growth. The development of the aircraft accelerated the demand for more mapping, and making it possible for photogrammetry (mapping from air photographs). The need for smaller-scale coverage of large areas such as the aeronautical charts, promoted large-scale mapping of the unknown areas. Furthermore, seen from the vantage point of an airplane in flight appears somewhat like a map, and thus the frequent air travelers developed interests in maps. Today, photogrammetric methods are extensively used in the preparation of nearly all published large-scale topographic maps.

The Union of Photography in the latter part of 19th century provided relatively inexpensive means of reproducing an image on almost any material. From then technical developments in the photolithographic and photoengraving fields were rapid and continuous, and became inextricably woven into the methods of cartography. Today, high-speed, multicolor lithographic offset presses are capable of handling any kind of cartographic problem. Modern map construction with its inscribing, open window negatives, photo lettering and so on is as different from the mapmaking of 50 years ago as is the automobile from the horse and buggy.

Probably the most profound changes in the field of cartography occurring at present are a result of the continuing development of electronic techniques associated with telecommunications, automation and computer applications. The hardware consisting of satellites, computers, digitizers, plotters, printers, lasers, and the cathode ray tube together with the associated software are literally revolutionising mapmaking. In many applications these remarkable mechanical and technical advances have made the solutions to old problems easier. Cartography as a body of theory and method for dealing with problems of recording, analyzing, and communicating geographical information in efficient mapmaking is clearly aided by the computers, remote sensing, electronic typography and the other innovations.

7.7 Key Words

Orient, this was the practice in the middle ages in which the farthest area that was known at all and, in which Paradise had traditionally been located. The term orient has since acquired a different meaning, that's the reader to understand the map by turning the indicated directions. On maps to orient means to arrange so that the map and the earth directions correspond or, alternatively so that the north is at the top.

Bathymetric maps, relief maps of the bed of the sea.

Orthogonal, right angled; pertaining to or depending upon the use of right angles or perpendiculars.

Software, the already established program and routines for accomplishing particular computer tasks.

Cathode ray tube, a vacuum tube in which slender beams of high-speed electrons are projected onto a

fluorescent screen to produce luminous spots.

7.8 *Exercise*

Define the following terms:

Compilation, (c) Lithography,

SECTION 8

8.0 *General Draftsmanship* 8.1 *Introduction*

This section introduces learners to the application of traditional, basic tools in the generation of lifework. Reference is made to the main instruments, materials and their usage. Scribing, and associated processes such as masking and stick-up are also introduced. Further, it is advised that cartographic laboratories and equipment be kept clean and tidy all times.

8.2 *Objectives*

- a) To introduce the basic instruments of draftsmanship
- b) To familiarize the learners to various drawing materials for cartography.
- c) To provide instructions for the maintenance of cartographic materials and tools.

8.3 *Introduction*

Drawings are produced from map compilations, which are generated as a result of research and the processing of available source materials. At this stage of map preparation each of the included symbols must be produced according to the definitive instructions

detailed on a specification list. This is because the efficient communication of spatial information by means of a map is completely dependent upon the quality of the graphic, i.e., it must be clear, legible, accurate, carefully produced and not open to misinterpretation or misunderstanding whilst also creating the impression. These factors can be enhanced or destroyed by the quality of the employed draftsmanship. Thus, for example, lifework must be precise in terms of width, length, and crispness; exhibit regular shape, orientation, intersections, and angularity at corners; be of appropriate density – sharpness and regularity; and allow the reader to distinguish between variations in width, must particularly where lines of different thickness connect one with another.

8.4 Methods of Map Drawing

(a) Pencil or Ink Drawing

Pencil lifework is normally employed during the preparation of original survey plots, whilst drawing pens and ink are used when particularly clear and well-defined lifework is required. The production of this is simpler, and results in graphic detail, is fit for immediate use, but requires considerable expertise. The application of tubular/reservoir pens with interchangeable points of varying thickness, or of technical pens, does not require the same amount of skill.

(b) Scribing and Mask Making

Scribing involves the engraving of lifework on the actinically opaque coating of a transparent, stable-based plastic film in order to produce a negative image. Its main advantage is that detail of a consistently high quality can be generated quickly and easily at the required scale, without the need to use the camera. There are two methods used for the creation of inscribing image-“wrong reading” (minor image) and “right reading”. The latter is the simpler to check against a compilation, but its use necessitates the employment of a special process during the manufacture of offset printing plates.

(c) Painting, Stick-up and Dry-Transfer

Either using a brush or an airbrush may carry out painting. It is used rather than peel- coating to produce masks for large areas destined to appear either as even tints, tints decreasing in density (vignettes), or as an overall grey-shading providing coverage of the complete area mapped. The size of brush used depends on both the extent of an area and complexity of its outline.

Stick-up (applied with wax or solvent) and dry-transfer (involves peeling from a backing sheet) are both terms applied to the removal of an image from a ready-made sheet of identical symbols (i.e., lettering, settlement symbols, etc.), and positioning it on an original drawing. The sheet may be commercially produced or made in a drawing office using a photo-typesetter or contact copier.

8.5 Instruments, Tools and Materials

- (i) The layout and the lighting of a drawing office
- (ii) Drawing board, whose size in millimeters depends on the height of the cartographer.
- (iii) Lighting involves 500 lumens emitted by fluorescent tubes, drawing board illuminated by white fluorescent tubes or an adjustable incandescent lamp, which can be moved to eliminate shadows, and lumens shining on the board, among other things.
- (iv) Drawing desk, which is normally in an inclined position, and the drawing board is in adjustable angle.
- (v) Desktop graphic workstation. Increasingly computerized workstations are becoming a feature of cartographic laboratories. They are employed for the interactive preparations of compilations and manuscript materials by using a keyboard and a mouse to manipulate displayed data.

(vi) Lighting Table. It is advisable that illumination of the table be provided.

Uniform illumination, together with some protection from the heat emitted from the lamp can be provided by incorporating (1) sheets of 5 mm thick glass which is matt on one side, (2) a sheet of milky-white plastic, (3) opal glass.

(vii) Other commonly used drawing materials using: waterproof ink include tracing paper, tracing cloth and cartridge paper; (b) suitable and stable to draw using plastic ink only are Astrafoil, Astralon and Durafilm.

8.6 Instruments and Tools for Drawing

Drawing instruments are tools that the draftsman uses include:

- 1) Drawing pens (Ruling), are used for constructing straight and curved lines.
- 2) Mapping pens are writing and drawing nibs. They are graded according to sizes, from those drawing very thin to thick lines. The choice for use of these pens is dependent upon the line gauge required to suit the task at hand. Although these pens are for different purposes, practice usually determines the choice of size that yields the best result for free hand lettering.
- 3) Graphos pens are a range of drawing and writing nibs of different gauges. The pens are not set to get the required gauges and they use a reservoir of waterproof ink.
- 4) Setsquares are used as guides to draw straight lines, or they can be used in pairs to set parallel or perpendicular lines. The setsquares that are commonly used are made of celluloid materials and they take the right-angled shape, although others are 45°/45° and 60°/30°.
- 5) Protractors are employed in laying angles, and for this reason their edges are graduated into degrees and half-degrees. They are made of transparent materials of either full-circle or semi-circle.
- 6) French Curves are use in drawing line around irregular curves. They are of various shapes and sizes. They are made of either celluloid or wooden materials.
- 7) Railway Curves are used in drawing arcs of constant radius and they are made of same materials as in the French curves.

- 8) Splines are flexible pieces of materials (wooden, metal, plastic) used as guides in drawing long curved lines, e.g., Meridians and parallels.
- 9) Drawing compasses are used in drawing circles or arcs. They have two legs hinged at the top by friction joint for establishing radius. One leg ends in a needle point and the other which is provided with a clamping screw and socket can accommodate a pen, a pencil, needle point or extension leg attachment.
- 10) Dividers are similar to compass, except their legs that are provided with needlepoints. They are used for laying a series of equal spaces and transfer of distances on maps and plans.
- 11) The Pantograph, useful for the reduction or enlargement of detail up to 6 or 7 times. In addition, it can be employed in the copying of drawings consisting essentially of a complex of curved lines.

8.7 Maintenance of Draftsman's Instruments and Materials

- a) Pencils should be kept sharp, with regular sharpening of lead, using a knife and a sand paper to bring them to a needlepoint.
- b) Pens, dividers, and compasses must be cleaned immediately after use, and their pencils must be protected against damage.
- c) Protractors, setsquares, scales and rulers must be kept clean and their measuring edges carefully protected from damage. Plastic setsquares, protractors and related instruments must be protected from heat and corrosive types of ink, e.g., plastic ink.
- d) Straight-edges, and parallel rulers if made of steel must be kept clean and free of rust.
- e) Drafting surface for cartographic work
In order to produce accurate maps, the materials and the surface on which the drafting is conducted must fulfill certain conditions, namely:
 - i) Dimensional stability, quality of material that withstands changes in temperature and humidity. This is an important requirement for maintenance of uniform scale.

- ii) Ink adherence, the ability of the drawing material surface to hold ink. iii)
Translucence, the ability of materials used for drawing purposes to facilitate reproduction and tracing.

Key Words

Navigation chart, detailed maps for guiding pilots and captains in the air and sea, respectively.

SECTION 9

9.0 LETTERING 9.1 Introduction

In this section the meaning of lettering and other related terminologies is provided. Freehand lettering, a common method used to communicate in languages applicable in mapping purpose, with the essential characters is also described in this section.

9.2 Objectives

- (a) To provide the meaning of lettering and other important terminologies used in lettering,
- (b) (b) To examine the methods used in the application of lettering.

9.3 Definition of Lettering

Lettering is a method of symbolizing and encoding details representing geographical reality. The cartographic quality and appearance of a map is dependent on the way in which lettering has been used. Distinction and recognition of a feature to which a name label applies, the search-time necessary to find names, and the ease with which

letters can be read are all important to the function of a map and its success or failure as a medium for the communication of spatial information to user.

9.2 Terminologies in Lettering

As in any graphic arts, there are some terms used in lettering and they are listed below:

- a) **Point Size.** This refers to the height of letters and specifically the capital letters. There are two point size systems namely: Didot and Pica. Didot system has approximately 3.75 points in a millimeter. Letters are referred to as point 12 or 20 point. The smallest point used in mapping is 4 point. The tendency at the moment, however, is to specify the size of lettering in millimeters, and 1.4mm is regarded as the smallest size that can be used in mapping. The Pica system is commonly used in Britain, whereas the Didot system is employed in other parts of Europe.
- b) **Upright Lettering.** Lettering that takes the vertical angle or 90 degrees to the baseline of the lettering.
- c) **Italic Lettering.** Letters slope towards right from the base line. The angle of the slope may vary from style to style but on average it is about 20 degrees from the vertical.
- d) **Bold Face.** This is lettering with thick strokes.
- e) **Medium Face.** This is lettering of medium thickness, the strokes being about a tenth of the height of lettering.
- f) **Light Face.** This lettering formed by light strokes.
- g) **Condensed Lettering.** Lettering in which the width of the letters is narrow compared with its height. It is used where a word is to be squeezed into a confined space on the map.
- h) **Style.** The manner of writing letters, which range from very simple styles to ornamental lettering. In cartography, relatively few styles are used and it is desirable that they are simple and legible. The choice of a style is part of map design.
- i) **Roman.** This is the name applied to Latin alphabet and is mostly used in European countries and American languages. It is also used to describe upright letters for Roman and sloping letters for Italics.

9.2.1 Freehand Lettering

This is the most flexible method available for the application of names to a map. Typically, the layout of titles, legends and so on will be required, and also the computation of letter spacing during the design process prior to obtaining of stick-up materials. Thus the need for individuals to know three vital aspects: spacing, the use of guidelines, and stroke technique. In the ensuing section, explanations are provided related to Gothic and Roman characters, which are the two most commonly used sorts of cartographic type styles.

Gothic, upper and lower case

Single-stroke lettering, consists of characters (Figure 9.0) in which the width of all line work is equivalent to the thickness of the point of the pen with which it is drawn. The method is used almost exclusively for letters, which are less than 8 mm in height. It is the easiest lettering style to produce and also the quickest.

Fig 9.0 Upper Case Letters: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Fig 9.0 (a) Italic Upper Case: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Lower cases are the small letters of the alphabet and are normally cm of the 0.6 height of the capital letters. Like upper case, freehand strokes are consisting of those techniques of drawing straight down, at an inclined angle, horizontally from left to right and curving from top to bottom. Up strokes cannot also be made with a sharp pen or pencil unless an aid such as a glass ruler is employed. These letters can be easily prepared in pencil with a plastic device such as the spacing template illustrated in Figure 9.1

Fig 9.1 Lower Case Letters: a b c d e f g h I j k l m n o p r s t u v w x y z

Fig 9.1 (a) Italic lower case: a b c d e f g h I j k l m n o p r s t u v

w x y z The following are some hints to help beginners to overcome some of the common difficulties in freehand lettering:

- a) Sit comfortably with your drawing at the most convenient position on the table.
- b) Draw fine sharp guidelines for your lettering in soft pencil. For lower case lettering guidelines indicate the waist height of the letters. Both horizontal and vertical guidelines must be parallel.
- c) Always keep a piece of paper on your drawing material on which to rest your hand. Thus avoiding soiling your drawing.
- d) To begin with always make an outline of the letters in pencil.
- e) Before beginning an upright lettering, it is advisable to draw lines vertical to horizontal guidelines. This is done at intervals in order to ensure that the length of words to be printed is covered.
- f) In sloping lettering, the same should be done with sloping lines at about 20 degrees from the vertical at intervals along the guidelines.
- g) Space letters so that lettering appears even, although the actual spacing between letters varies. For instance, there should be shorter distance between 00 and 11. Between words leave a space of approximate width of letter o.
- h) Letters are formed by several strokes and each stroke should be clearly defined. First, place the arm in a resting position above the supposed end of the stroke and mark the point. Stretch the hand a little beyond the beginning of the stroke and aim to bring down your pen in the proper direction.
- i) Make strokes with the motion of the full arm (not with the fingers) and wrist only, “follow through” lifting your pen in the same direction as the pen.
- j) Curved letters should be formed from two to four segments. Do not go over the same stroke twice.
- k) Letters smaller than 2.5 cm in height should not be ruled with ruling pen and straight-edge because the result would be uneven lettering due to inaccuracies in “weight” of the strokes.
- l) Always keep your pen clean.

Name Placement

Lettering placement should express itself by style, nature of the feature, size, extent of the feature, spacing, importance of the feature, and placement.

Examples:

- 1) Style, ornamental lettering found in most certificates.
- 2) Size, large lettering as found in titles of maps.
- 3) Placement, positioning of the names should be done in such a way that it does not clash with other map details. Note that there are 6 positions where to place the name without masking other map details.
- 4) Spacing, lettering in annotating of an elongated feature, e.g. a river or a mountain range.

SECTION 10

10.0 MAP SCALES AND PLOTTABLE ERROR 10.0 Introduction

The meaning and use of map scale are described in this section. The scale conversion and units of measurement of scale are also provided. The types of scale representations namely: graphic, written statement and fraction, and the significance of conventionalizing the features on the map, that in topographic mapping are representing the earth as faithfully as possible within the limitations imposed by the map scale are in addition presented.

10.1 Objectives

- a) To describe map scale and how it is used in map compilation and evaluation of map accuracy.
- b) To appreciate the significance of symbolizing or conventionalizing the features on the map.

10.2 Map Scale

The scale of a map is the ratio between any distance measured on the map and the same distance measured on the ground. One map is said to be of larger scale than

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another when a given distance on the ground is represented by a larger measurement on the map. Conversely, on a smaller scale map, the distance is represented on the map by a smaller measurement.

Example

On map A, 1 km on the ground is represented by 1 cm on the map.

On map B, 1 km on the ground is represented by 1 mm on the map. Map A is at a larger scale than map B. Mistakes are often made by thinking that since map B covers a larger area than map A, therefore has a larger scale. In general, it can be concluded that the more “miles to the Inch”, or “kilometers to the centimeters”, the smaller the scale of the map.

Units of Measurement

The standard unit of measurement for all horizontal distances in military land maps is in meters. All scales and grids are therefore in metres, except in certain obsolescent maps, where many are still in miles, yards and feet. The unit of height measurement is also in metres. On aeronautical charts, the horizontal unit is in nautical mile.

There are three methods of representing scales:

- 1) Representative Fraction (R.F.), e.g., 1:50,000 or 1/50,000. 1:50,000, means that one unit on the map represents 50,000 units on the ground. The relationship expressed by the function remains unaltered whichever the unit of measurement is used. For example, 1 cm on the map represents 50,000 cm on the ground or 500 metres.

- 2) Written Statement

On maps, which use other units of measurement, rather than the metric system, notably the British maps and navigational charts it is common to use statement of scale. For example, one inch to a mile, or three nautical mile to an inch.

- 3) Graphical Scale (Bar scale)

This is an accurately divided line from which distances taken from the map may be read off as distances on the ground as shown in figure 10.0. There are two types of linear scale, namely; single line and double line dived.

They are constructed to show:

- (i) Kilometres and 100s of metres,

- (ii) Miles and furlongs,
- (iii) Nautical miles and cables in case of nautical navigation charts. What is shown on a scale line depends on the scale of the map; say 1:1,000,000 would show 10s of kilometres and kilometres.

For one to determine the length of the scale, the following formula is used:

$S = MD/GD$, where S is a scale shown as a function, e.g., 1/25,000

MD is the map distance

GD is the ground distance.

Example

At a scale of 1:25,000, how long would the scale line be showing 5 km?

$S = 1/25,000$; $GD = 5$ km. The MD will thus be calculated as:

$$\begin{aligned} S &= MD/GD, MD = S \times GD \\ &= 1 \times 5 \times 100,000/25,000 = \\ &20\text{cm.} \end{aligned}$$

The length of the scale line therefore would be 20 cm.

10.3 Plottable Error

Plottable error is an unavoidable inaccuracy, which is manifested in all plotting on maps and its magnitude is dependent upon the map scale, i.e. the larger the scale the smaller the plottable error, and vice-versa.

For the purpose of map evaluation, it is based on the theory that the finest line one can draw unaided is 0.01 of an inch in thickness. Note that the scale of a map is defined as a ratio between measured distance on the map and the corresponding actual distance on the ground. Thus: Map scale = Map distance (MD)/ Ground distance (GD).

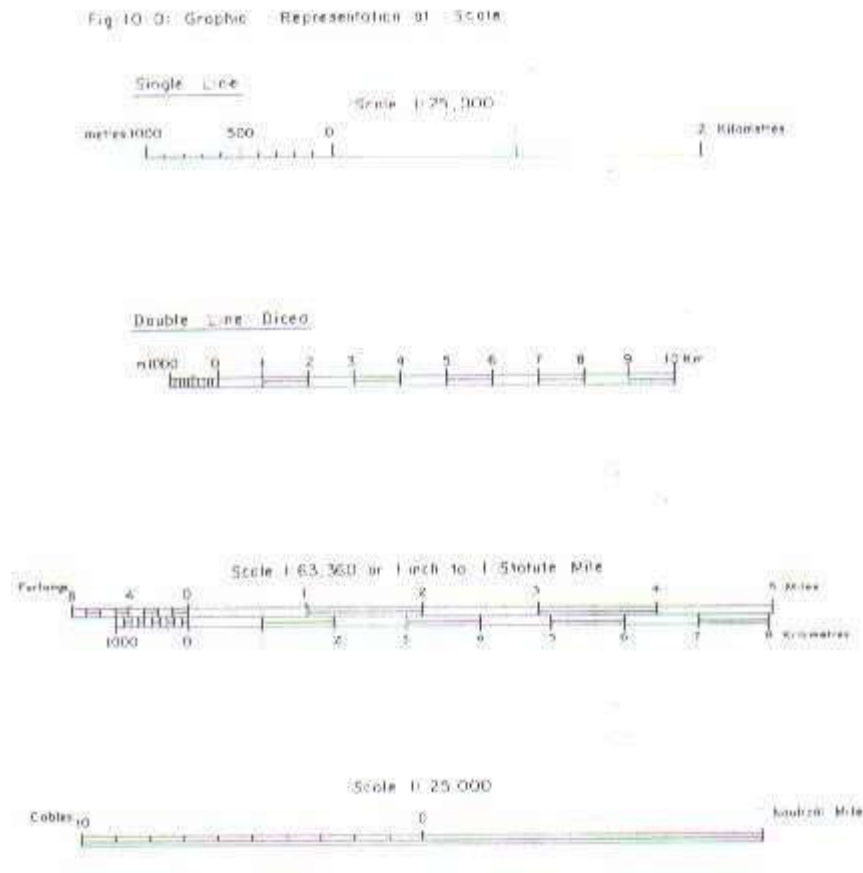


Figure 10.0 Graphic Representation Scale

Example

Calculate the plottable error for a map at a scale 1:50000

Map scale = MD/GD = 1/50000

Note that the finest line one can draw unaided equals 0.01 of an inch

Thus the equivalent of 0.01 of an inch in terms of ground measurement at scale 1:50000 is

Scale = MD/GD = 0.01/GD,

1/50000 = 0.01/GD

GD = 0.01 x 50000 = 500 inches = 500/12 = 41.76 ft.

This means that a length of 0.01 inch on the map of scale 1:50000 will be representing approximately 42 ft of the ground measurement. The 42 ft is thus the plottable error of any at the scale 1:50000.

Table 10.1: Plottable Errors at Various Scales Commonly Used in Mapping:

Scale	1:12500	1:25000	1:50000	1:100000	1:125000	1:1000000
Plottable Error	10.5 ft	21 ft	42 ft	84 ft	210 ft	840 ft

It is clear from Table 10.1 that the magnitude of error is inversely proportional to the map scale. In map designing, the purpose of the map determines the map scale. Hence in map compilation, the map scale is pre-determined. When compiling a new map, details to be incorporated are enumerated, and a detailed appraisal is prepared. Guided by the plottable error of the chosen scale, some details are either symbolized or conventionalized, generalized, simplified or omitted altogether depending on their significance. Details that are according to their dimensions, either within or near the plottable error cannot be plotted to scale. For example, a church building whose dimensions are 100ft by 42 ft at a scale of 1:50000 cannot be plotted according to its planimetric layout because one side is smaller than the plottable error of 42 ft. However, given that a church is an important landmark institution, it needs to be included in the map. The cartographer has to use the intuition and discretion to devise a method of representing it, and thus the employment of symbols to represent features such as churches, mosques, schools, and escarpments, among others on the map.

On a scale map of 1:12500 whose plottable error is 10.5 ft will be easy to plot the church with the above planimetric layout. It should therefore be observed that as the scale of the map becomes smaller, more symbols are used to represent features on the ground. Conversely, large scales facilitate the plotting of features according to their planimetric layout and enhancing accuracy.

10.3.1 Summary

Plottable error enables the cartographer to accomplish the following:

- 1) To decide which feature(s) to be omitted if found not to be significant according to the purpose of the map.
- 2) To determine which features are too small in their dimensions, but are significant according to the purpose of the map, therefore to be symbolized or conventionalized.
- 3) To include all features that are large enough to be plotted to scale.

10.4 Key Words

Simplification. Features, which are too small or too complex to be plotted on the map using the same scale. But they are simplified to enhance eligibility, clarity and avoid overcrowding by symbols.

Conventionalize. Using symbols to plot details according to their planimetric layout.

10.5 Practical Exercise

Using a topographic map scale 1:50,000 covering your local area:

- a) List down the features that you can identify on the ground and are represented on the map by the symbols.
- b) Features plotted to scale on the map. In order to make proper judgment, identify the features on the map and take the actual measurements on the ground for the same.
- c) Measure the width of one major road, and determine if it is plotted to scale. Justify your answer.

SECTION 11

11.0 Map Specifications 11.1 Objectives

- a) To familiarize with features incorporated in topographic maps, their drawing, colors and sizes.
- b) To understand the need for adhering to technical instructions for the maps to be uniform.

11.2 Introduction

Map specifications can be regarded as the instructions given to the cartographer about the way in which all the features are represented on the map.

These instructions include:

- 1) Map frame, where the map is framed using single line or multiple line borders, as shown in figure 11.0.

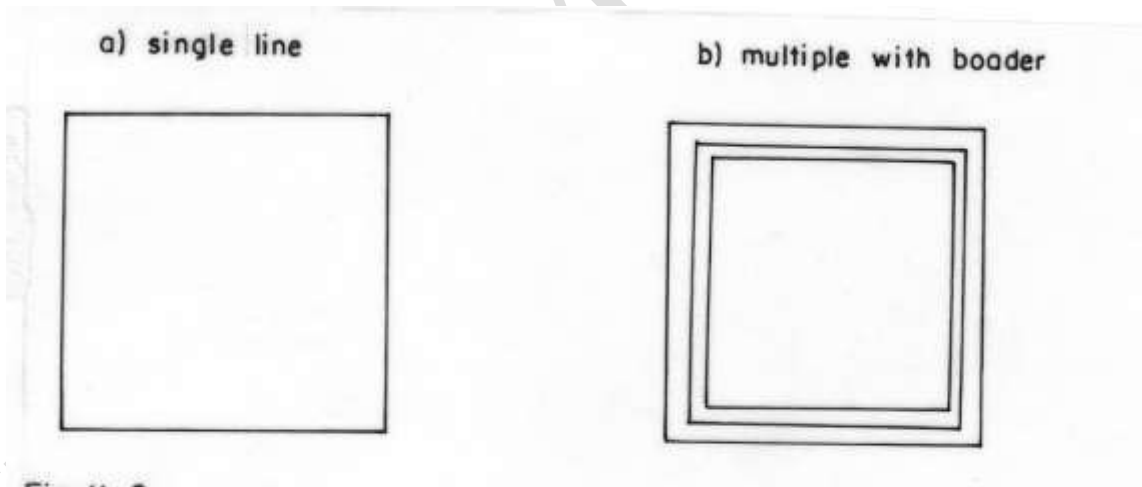


Fig. 11.0

- a) single line
- (b) multiple with border
- 2) No frame or a floating map whose map face ends with the paper on which the map is drawn. This is the map without a neat line.
- 3) Map Title. Usually located outside the frame at the top center of the map, and has fairly large upright letters.

- 4) Map Scale. This can be in the form of statement scale and line scale (bar scale) and are together in the same map. For instance, scale 1:50,000 can be placed slightly outside the frame at the bottom center for series maps, or where there is space in other maps.
- 5) Geographic values (Graticule). They are shown at convenient intervals around the edge of the sheet.
- 6) Legend. This is a collection of explanatory words on a map. Note that without a legend information cannot be extracted from the maps. The legend is usually placed at the bottom margin of the map.
- 7) North Points. These are True, Grid, and Magnetic Norths in their relative locations. They can be placed at the bottom margin near the legend, either left or right.
- 8) Printer's Note. Placed at right side of the bottom margin.
- 9) Series Number and Sheet Number. Placed at the upright margin of the map. The series is normally repeated at the left bottom margin.

- 10) Compilation Note. Located at the bottom margin below the Series Number.
- 11) Copy Right. Sole legal right held by the compiler of the map. Situated at the left margin below the Compilation Number.
- 12) Other important features are Index Diagram, Boundary Diagram, Reliability Diagram, and Grid Reference Box, all located at the bottom margin of the map.
- 13) Contents of the Map which fall under the following headings:
 - a) Point Features, e.g., cities with 5,000,000 inhabitants are represented by 1.5mm square of black and red infill; towns with more than 200,000 inhabitants are represented with 1.5mm radius circle of black and red infill; and towns with less than 50,000 inhabitants has 1.5 mm radius with no infill. Villages are represented with 1 mm solid black, and airport and airfield are represented with 2 mm red solid arrow and 1.5 mm red solid cross.

11.3 Key Words

Graticule. Network of meridians and parallels, normally used in small and medium maps.

Conventional Signs. These are internationally understood and accepted symbols. For example, water is signified with blue color, green signifies vegetation, and black cultural features, among others. **Map**

Symbols. Maps drawn for internal use can employ any sign or color to represent any individual details of a feature provided it does not cause any confusion. Hence conventional signs are symbols, but symbols are not necessary conventional signs.

11.4 Practical Exercise

- 1) Using 1:50,000 Topographic Map, identify all the items listed 1 to13 in section 11.3 above, determine their relative location on the map frame, sizes, and colors, among other features. Comment on their appropriateness and if they require any improvement.

- 2) Study the map marginal information and the conventional signs shown in the legend. Comment on whether the layout of the legend needs some improvement.
- 3) Series Number and Sheet Number are shown at the top right and repeated at the bottom left. Why do you think this is necessary?

SECTION 12

12.0 Portraying Land-Surface Form 12.1 Introduction

The portraying of land-surface form is in reality depiction of relief on maps. The representation of relief on a map with elevation and shapes of the ground, either above or below a datum (reference level) employs many techniques such as benchmarks, spot heights, contours, and hachure among others. The datum is usually the mean sea level.

12.3 Objectives

- a) To introduce various methods of depicting relief,
- b) To explain why contour method is regarded as a standard method,
- c) To categorize the methods of depicting relief.

12.4 Introduction

Relief is a general term applied to the shape of the ground in a vertical plane. The representation of relief on a map is depicted with heights and shapes of the ground, either above or below a datum. The datum is normally the mean sea level. The depiction of depths below sea or water level is called *bathymetry*.

12.5 Methods of Depicting Relief

There are two distinct elements in the representation of relief, namely:

- a) Height Representation, and
- b) Shape Representation.

Representation of Height is a factual matter in which the variations will arise from the type, density and accuracy of the information required. On the other hand, representation by the shape may be largely artistic, and there is much scope for variation according to taste. It is often the combination of these factors, which determines the appearance of a map in any area where there is a marked relief.

Bench Marks

These are positions of permanent marks, usually depicted by a symbol and the height recorded besides it. The most precise Bench Marks are those derived directly from a leveling network. On maps of 1:10,000 and larger scales, the values may be shown to or more decimal places. On 1:25,000 scale maps, the values may be either omitted or rounded off. On smaller scales, Bench Marks are unlikely to be shown except where they are very useful marks such as Spot Heights. Bench Marks are not normally shown on military maps, unless they are already shown on maps adapted for military uses.

Spot Height

This is a height provided for a point marked by a dot on the map in order to supplement the information given by the contours. Spot heights are best placed at ruling points such as tops of hills or slopes, bottom of valleys, ridge points and saddles. Their choice and placing is important and they should be sited to give maximum value in determining the height of the ground where this is not clear from the contours or other information. This is particularly significant for those who rely on aeronautical charts.

Trigonometrical Height

This is a pillar or other recognizable mark defined on the ground as a form of survey control point. The heights of such points are normally given besides any of the symbols in figure 12.0 for the point.

☐ Primary trigonometrical station

☐ Secondary trigonometrical station

□ Other stations

Figure 12.0 Commonly used Symbols to Depict Trigonometrical Height.

Contours

A contour is a line joining points of equal height as depicted in figure 12.1, and is the standard method of relief depiction. Contouring combines an accurate indication of height with a good measure of shape, and is a simple and effective method of showing relief, especially when used in conjunction with heightened points such as spot heights.

Fig. 12.1 Contour Showing a Hill with a Spot Height Marking the Top.

The effectiveness of contours depends to a large extent on the selection of the best vertical interval between contours for the scale of the map and the nature on the ground. The best contour interval is one which enables the hill features and changes of slope to be shown as fully as the scale permits without making the contours too close together to obscure the map, as usually shown on the steep slopes.

Owing to the differences in types of countries and purposes for which the map may be required, it is not possible to lay a standard vertical interval. For this reason, the following is a general guide for a normal topographical map of an average relief:

<i>Scale</i>	<i>Vertical Interval</i>	
	(Meters)	(Feet)

1:10,000	5 (or 10)	25
1:25,000	10	25
1:50,000	20	50
1:100,000	25	100
1:250,000	50	200

On smaller scales, it is necessary to show the 50 Meter, and 100 Meter contours to vary in an interval of higher ranges.

Contours are drawn as continuous line of a constant gauge. Every contour is called an “Index Contour,” and is shown in a heavier line gauge. This helps in reading and counting of the contours to determine height contour values placed on breaks made on the contour lines as shown in Figure 1, and positioned so as to be read when the map user is looking up the slope. They are often placed in “ladders”, one above the other to facilitate reading, but this is neither always possible nor desirable.

The accuracy of contours will depend upon the method and accuracy of the survey. On large-scale maps, accuracy is determined by leveling and is normally quite accurate. On small-scale maps, the accuracy is less precise, and they only provide a more generalized picture of the ground level and should be recognized as such.

Laying (Altitude Tint or Hypsometric Coloring)

A layer is a uniform tint applied to all ground between defined limits of height above or below a datum, e.g., all ground above 50 Meter and 100meter above sea level. By using different tints for different layers, it thus easy to give a clear picture of variations in height or depth over an area. Layers are normally used in conjunction with contours to assist the user in gaining a quick appreciation of relief. They are occasionally used alone or in conjunction with **hill shading** to give general impression of relief in areas where there is insufficient accurate height information for contouring. Although layers are considered to be an extremely useful aid to relief appreciation, they involve

considerable additional work in cartography and printing. They are thus used only on mapping at scales of 1:250,000 and smaller scales.

The problem with layering lies with the selection of colors for the different layers to satisfy the following requirements:

- a) There must be a clear distinction between adjacent layers,
- b) A pleasing gradation of colors,
- c) A range of colors which will satisfy all changes in elevation over the map series,
- d) The tints must not be so strong as to obscure the map details and names,
- e) The tints should desirably not produce the effect of a clash with nature, e.g., green tints in a desert area,
- f) Where tints are used for other purposes, e.g., for woods, the layer tints must permit these to be distinguishable.
- g) The different tints required should be capable of production in minimum number of color printings to save cost and time.

Fig. 12.3 Layering of Contour marking the beginning and ending of each color band.

Hill Shading

This is the most commonly used current technique to indicate shape, either alone or in conjunction with contours and/or layers. It does not itself give any indication of height. It is only steepness of the slope, which is depicted by the Hill shading, but it gives an excellent visual picture of relief without any positive value of height.

Basically, Hill shading is consisting of the “dark” side of a hill and lighting up of a sunny “side” to provide contrast; the darker shading the steeper the slope on the shadow side. The light is assumed to come from the North West corner of the map. For some purposes a good hill shaded map is more descriptive than a contoured map. With the addition of spot heights, a reasonable standard of height comparison may be obtained and a clearer picture of general relief is shown. It cannot, however, be a

substitute for a contoured map, where positive height information is necessary. A disadvantage of hill shading is that it tends to obscure names and details below it, and for this reason it has to be used more lightly than might otherwise be desired.

Form Lines (Figure 12.3)

These are approximate contours sketched to show the general shape of the terrain rather than the elevation. They are used when it is not possible to obtain accurate measurements of contours, a situation which may occur for part or for complete map sheet. They can be used in conjunction with contours and Spot Heights to show the rough shape of the terrain in inaccessible areas, or built areas. They are usually depicted by broken lines, but are not given height values, and they should not be mistaken to be contours.

Fig. 12.4 Form Lines

Hachures

They depict relief by means of short disconnected lines drawn down the slope in the direction of water flow as shown in figure 12.5. The lines are short and close together on the steep slopes, and longer and more spaced out on the gentle slopes. This is an artistic method which requires much skill, although it does give a three dimensional effect to relief. However, it does not provide any definite height information, and has the disadvantage of obscuring other details. This method resembles the pictorial type of relief depiction used on some of the earliest maps. Hachures are not normally used on modern mapping, except to depict cuttings, embankments as steep slopes.

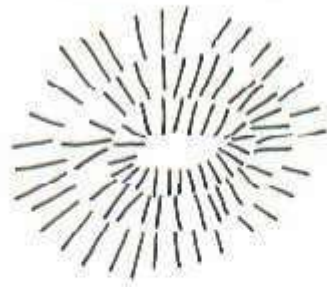


Fig 12-5: Use of Hachure to Depict a Hill

Bathymetric Relief

This the depiction of depths below sea or water level. Bathymetric relief when required on land maps is depicted in a similar way to ground relief, viz: by depth values and contours. They are shown similarly to ground contours, except that they are depicted in blue or black. Their values are also related to sea level, but in inland waters, they are related to surface level of the water. The information in coastal areas is normally obtained from *Admiralty Charts*. The unit of measurement in new charts will show depths in fathoms or in feet. (One fathom equals to six feet).

Plastic Relief Maps

They give an excellent depiction of relief as it takes the form of a model with all other details printed on the surface of the model. In order to produce the map, a model of the relief is made in addition to the normal detailed drawings. Moulds are then made of the model. The details are printed in a conventional manner on to a plastic medium, which is inserted into a mould and placed in a steam press.

This type of relief depiction is not suitable for most tactical military purposes since bearings and distances cannot be plotted and the map cannot fold. However, it is important for strategic planning work.

Relief Ornamentation

These are certain relief details which cannot be shown by contours and for which symbols are necessary. Such relief items include cliffs, embankments and cuttings, rocks and boulders, and sand features. These symbols are normally shown in the same color as contours (usually brown), altogether in some cases they are shown in black. Unless there is a special technical requirement to such items in detail, the simplest form of symbol should be used to indicate the existence and extent of the item.

In high mountain areas, glaciers and allied features must be shown by special symbolization. In areas of perpetual snow, contours are usually shown in blue, with exposed rock formations in black.

12.6 Practical Exercise

Using a topographic map scale 1:50,000 carry out the following exercise:

- a) List down the methods used to show relief in the whole map.
- b) Using contours, identify the highest and lowest regions on the map.
- c) Using illustrations and guided by the topographic map, show how you would depict the following features on a map: (i) Outcrop rock (ii) Boulder rock (iii) Valley and a spur (iv) Steep slope (v) Quarry (vi) Crater.
- d) Using contour lines and values, draw the following features:
 - (i) River Valley
 - (ii) Hill
 - (iii) Mountain Range
 - (iv) Index Contour.

PART THREE SECTION 14

14.0 MAP ANALYSIS 14.1 Introduction

This is an introductory section for Part Three, Map analysis, which highlights the most important aspects of maps such as drainage basin, settlement patterns and network analyses. In this introduction, modern maps produced by traditional methods with assistance from computers or, as is becoming increasingly common, by a combination of both conventional and automated techniques are studied. In all cases, it is essential the learners familiarize themselves with the definition and uses of maps. It is also important for the learners to understand the methods of studying and language of maps.

14.2 Objectives

- (a) To provide the definition and uses of maps.
- (b) To explain the methods and language employed in the study of maps.

14.3 Definition of a Map

A map is defined as a graphic representation of the whole Earth or a position of the Earth surface, or any celestial body drawn on a piece of paper at any scale.

14.3.1 Uses of Maps

Generally, geographers use maps for:

- (a) Providing information such as distribution of phenomena. Such phenomena may include population, settlement, and hydrographical features. Maps depict the location of these features and their relative pattern of distribution.
- (b) Discerning spatial relationships between various phenomena; i.e., gaining insights into relationships of various features. For example, relating vegetation to altitude and human activities.
- (c) Making descriptive and statistical analysis. Maps form a basis for systematic description of a given area and can classify descriptions by offering geographic and pictorial illustrations.
- (d) Identifying geographical problems for study purposes.

14.3.2 Study of Maps

There are three stages involved in the study of maps. These are:

- (1) Map Reading, involves finding out which features are chosen on the map and where they occur.
- (2) Map Interpretation, refers to taking some decisions on why certain features occur where they do. Note that map reading must precede map interpretation.
- (3) Map Analysis, includes such aspects as reading and interpretation, plus an evaluation of the interrelationships of the various features on the map with the view to classifying and assessing them. Map analysis does sometimes imply some quantitative assessment.

14.3.3 Language of Maps

Reading maps requires an understanding of the language employed by makers of maps in producing them. The language can be studied in 4 broad terms: (i) Marginal information (ii) Conventional Signs and Symbols (iii) Scale, and (iv) Representation of Relief and Cultural Features.

Marginal Information

Maps usually have margins within which various kinds of information are provided, and any map-reader should first look at and understand such information. Marginal information will normally include the following:

(1) Title of the map

The purpose of the title is to indicate the extent the reader may expect from the map, or whether it is treating a general or a specific top. Maps may have broad titles such as „Nairobi and its environs“ or specific titles such as „Land Use in Kiambu District“.

(2) Date of Compilation, Surveying, Printing, Production and Edition

The information helps the reader to estimate the reliability of the facts contained in the map.

Nationality, Grid References, Map References, Magnetic Points and Variations This information helps the readers to locate the area covered by the map and also orient the map.

(3) Scale

This is important marginal information for calculations and measurements on the maps.

Scales are usually shown on maps as follows: (a) Representative Fraction (R.F.), (b) Statement scale, and (c) Linear scale (Refer to the details in lesson 10).

This information allows international comparability and ensures that readers in most countries can make use of the maps.

(4) Legend (Key to Symbols)

Makers of maps, use symbols to convey most of the direct information. Non-conventional symbols peculiar to a map should also be noted to facilitate exploration of the map.

Maps and diagrams give a pictorial view of spatial distributions and relationships. In order to show these characteristics of phenomenon, makers of maps employ conventional signs

and symbols. The symbols and signs used are supposed to satisfy four main requirements to facilitate adequate study of features on maps:

- (i) Each symbol must be uniform throughout a map or a series of maps,
- (ii) The symbol must be comprehensible, without leading to guess work,
- (iii) It must be legible,
- (iv) It must be sufficiently precise.

In various maps, conventional signs have been devised for depicting relief, water and drainage (hydrography), vegetation, communication, settlement, place names, boundaries and other miscellaneous features.

- (a) Relief includes such features as hills, valleys, slope types and plains, through contours, spot height, trigonometrical points, layer colouring, hachure, and oblique hill shading.
- (b) Hydrography consists of seas, lakes, rivers, reservoirs and artificial drainage canals and channels (aqua-ducts).
- (c) Vegetation comprises woodlands, high and low forests, conifers and deciduous forests, grasslands, parks, orchards, marsh, and rough pastures.
- (d) Communication include railways, roads of different classes, footpaths, ferry routes, water pipes and electric cables.
- (e) Settlement consists of rural and urban, which is also divided into large and small towns, large villages and small villages, hamlets, camps and public buildings.
- (f) Place names are in roman, Italics, upper and lower cases, and other various styles of lettering, letter sizes and shapes.

SECTION 15

15.0 DRAINAGE BASIN ANALYSIS 15.1 Introduction

This section introduces the learners to drainage basin analysis, the description of river systems, and stream ordering. Further, it deals with the advantages of quantitative description of river systems and the qualitative approach of river systems; the former exposing the learners to some measurements, which make it possible to describe or express drainage system in numerical terms.

One major concern of map analysis is the delimitation of physiographic regions, that is the dissection of topographical maps into component parts, for purposes of description, analysis and interpretation. Drainage pattern analysis is an important aspect of geomorphic studies. It deals with surface drainage interpretation.

15.2 Description of River System

A river system occupies a basin or a catchment area, which may be defined as the entire area of slopes leading into and drained by a river and its tributaries. Every stream and river system is separated from its neighbors by a watershed, water parting or divide, which is the elevated land separating the headwater's tributary to one stream or system from those tributaries to another stream or system. The water parting after has a highly irregular course, and although following the highest land, it does not necessary follow the highest ridge or a range of hills.

The quantitative study of drainage features on maps is usually made of three major headings or elements of the landscape, namely;

- (1) Linear properties (2) Areal properties, and (3) Relief properties.

Linear Properties

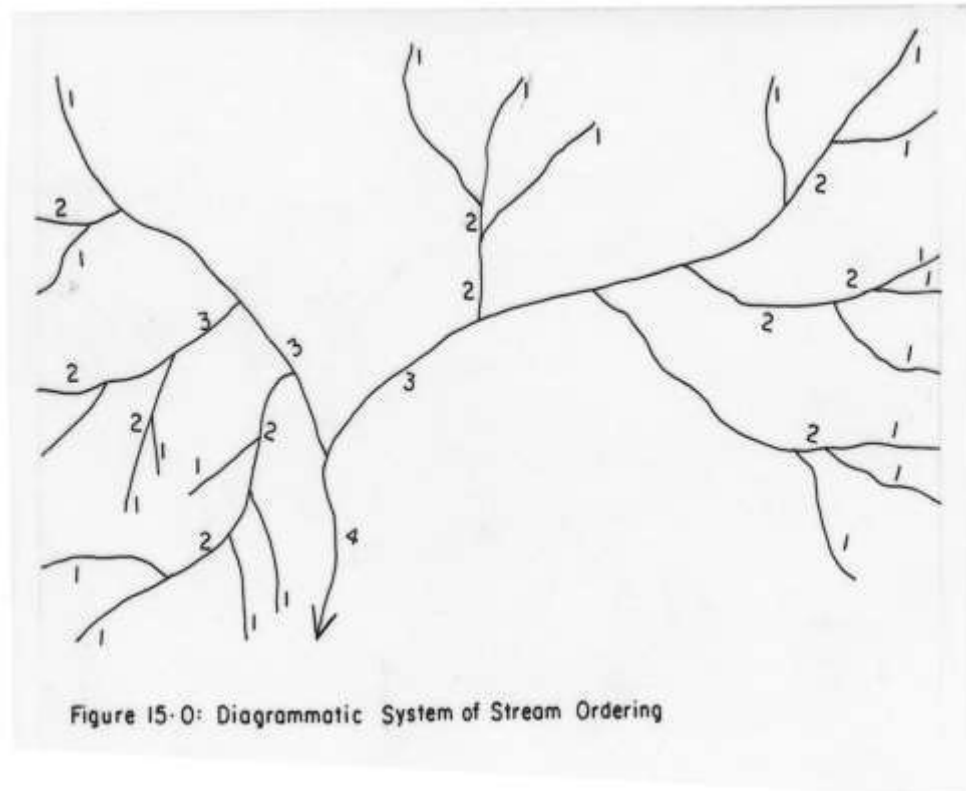
They refer to the stream and stream channels. The study of linear properties is therefore restricted to lengths, numbers and arrangements projected on a horizontal planimetric plane. For a long time, before 1945 drainage lines and segments were studied qualitatively and descriptively in the tradition of progress (William Davis 1945). Drainage lines were described according to their overall pattern and planimetric layout as dendritic, annular, trellis, rectangular, centripetal and radial, among other patterns reflecting mostly the lithology and the stage of their evolution (in the cyclic sense). Stage and anthrop-geomorphic terms such as youth, mature and senile (old) age were also used. Strahler and Chorley have extensively documented the limitations of qualitative and descriptive approaches to the study of river system.

The above terms are not readily quantifiable and so do not facilitate scientific study. Any assessment on these lines is also largely subjective and of limited use in comparing one drainage pattern with another. A recent development has been that of quantitative measurement. It is now possible to measure and express in numerical terms the characteristics of a drainage system. This has brought a great advantage to the river study, for it enables researchers in different regions to make quantitative or mathematical comparisons.

Horton (1952) is credited for revolutionalising the study of fluvial geomorphology. He applied statistical treatment to drainage nets. His concern was to enumerate the number and lengths of streams and investigate the relationship between the numbers and the lengths to the actual drainage area and with the density of drainage nets. Horton introduced among other things an analytical approach, which makes it possible to derive a number of mutually compatible dimensional and dimensionless parameters for drainage basins and drainage nets wherever they occur. The most important of these, which can be studied from topographical map; are stream order, stream length, bifurcation ratio, drainage density, stream frequency and drainage intensity.

Stream Ordering

Horton introduced the idea of orders of streams as a means of describing river systems. In stream ordering exercises, the mapped stream channel network of an area is divided into individual lengths of channels as channel segments, and arranged according to a hierarchy or magnitude of orders. There are many ways of stream ordering, but the one proposed by Strahler (1952), a modification of Horton's (1945) is the most popular. According to Strahler-Horton stream ordering system, each un-branched fingertip channel is designated as a segment of the **first order**. Two first order segments join to form **second order stream**. This second order magnitude extends down to the point where it joins another second order stream, where upon a third order stream starts, and so forth (Figure 15.0). The simple rule is that when two streams of the same order meet, they form a stream of the next higher order. A stream receiving another stream of a lower order than itself is not raised to the next higher order. For example, if a fourth order stream is joined by a stream of lower order such as 1, 2, or 3, it still remains a fourth order stream.



It should be noted that the application of this stream order concept along with statistical analysis of drainage nets is proving very fruitful in the study of rivers, their behavior, work and the landscape patterns, which emerge.

Bifurcation Ratio

The number of segments in each stream order can be counted and their various lengths measured. The ratio of the number of segment of a given stream order to that of the segment in the higher order gives the bifurcation ratio (R_b) expressed by the formula:

$$R_b = N_u / N_{u+1}$$

Where u is the order designation of a stream.

N is the total number of segment of that particular order.

The bifurcation ratio is calculated within pairs of the stream order, e.g., between first and second order; second and third order, etc.

Example

The total number of stream segments in a given basin is as follows:

$N_1 = 80$, First order

$N_2 = 40$, second order

$N_3 = 15$, Third order.

The bifurcation ratio can be computed as follows:

a) $R_b = N_u / N_{u+1} = 80/40+1 = 1.95 \approx 2.0$

b) $R_b = \frac{80}{40+1} = 2.67$

On average, the bifurcation ratio ranges between 2 and 4. However, in complex drainage basins, higher ratios have been recorded.

Horton's Law of Stream Numbers

The law states: „the number of stream segments of successively lower order in a given basin tend to form a geometric series, beginning with a single segment of the highest order and increasing according to a constant bifurcation ratio.“ That is to say, if in a given basin the highest stream order is sixth, then Horton's law of stream numbers is as follows:

Order sixth, fifth, fourth, third, second, first.

Stream No. 1 : 2 : 4 : 8 : 16 : 32

10 : 20 : 40 : 80 : 160 : 320.

Note that in the above basin the R_b is a constant of 2.

Interpretation of R_b

Low values suggest few streams of that order in relation to the next one than expected and that there is low level of stream development. High values suggest comparatively high number of streams of that order in relation to the next one, and shows considerable stream development.

15.3 Other Measurements of Drainage Basin

There are a number of other measurements about a drainage basin and they include:

- a) The area of drainage basin in KM²
- b) The maximum elevation of a drainage basin, H max., usually measured in meters.
- c) The minimum elevation of a drainage basin, H min., normally measured in meters.
- d) Relief of a basin is usually measured in meters. The relief of a basin is the difference between the maximum and the minimum elevation of a basin. Relief = H max. – H min. i.e., if the max. H is 5240m and the min. H is 3420m, then the relief of the basin is 5240 – 3420 = 1820m.
- e) The total length of stream channels in the basin, measured in km.
- f) Drainage density of the basin, usually measured in km/km².

Among these measurements, drainage density is the most important.

Drainage Density

Drainage density may be defined as the ratio of the total length of the streams within a single river system to the area drained by that system, and is calculated by dividing the total length of all the streams by the area drained:

Drainage Density (D.D) = total length of all the streams (km) ÷ area drained (basin area) in Km². The D.D is in Km/Km².

In this way drainage density can be defined numerically and this allows comparison to be made between areas. For example, it becomes possible to compare the drainage density between a very rainy area and a very dry region, or between areas of permeable and impermeable rocks.

15.4 Factors Influencing Drainage Density

The drainage density of an area is influenced by a variety of factors and conditions; the most important being: (i) rock type (ii) soil (iii) slope (topography) (iv) climate and (v) vegetation.

The relative spacing of drainage lines sometimes referred to, as drainage texture will obviously be affected by the permeability of the rocks, porous rocks such as chalk and limestone are highly permeable, and will militate against the development of a closely spaced drainage network. Conversely areas of clay with fine texture tend to have well developed drainage network. Climate has a greater control over drainage density. In arid regions there will be insufficient moisture to sustain permanent streams, whilst in areas of high rainfall are likely to result in an abundance of surface drainage line.

Exercise

1. Using examples of your own choice, show how vegetation and slope can influence the drainage density.

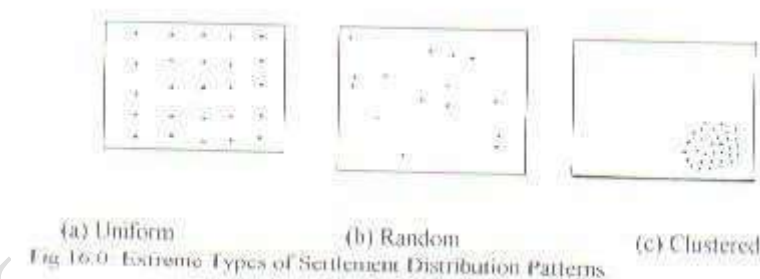
SECTION16

16.0 ANALYSIS OF SETTLEMENT PATTERNS 16.1 Introduction

Essentially, geography is concerned with distributions in space and, one of the most important distributions the geographer has to consider, is that of human settlements. For many years the pattern of population distribution tended to be described as either dense or sparse, whereas settlement was described as either nucleated, regular (uniform), linear, or random. This mode of description was merely qualitative. Recently, considerable attention has been given to providing a more precise mathematical way with which areal distributions can be described.

Basically, three extreme types of distributions can be distinguished, namely:

- (a) Uniform or regular (b) Random (c) Clustered (nucleated and aggregated) as shown in the figure 16.0, below:



The points may represent settlements (villages, schools, hospitals, churches, towns market places, etc). If the pattern is uniform, the distance between one point and its nearest neighbor will be approximately the same. There will be one or more groups with a relatively short distance between each point and its nearest neighbor, and large areas in which no point is located if the pattern is clustered. If the pattern is random, a situation arises in which there is some tendency to cluster, but also a number of points fairly widely distributed.

The problem was to find a single index for any given pattern, running on a continuous scale, from one extreme, a situation in which all points are distributed uniformly over the

whole area. One solution, originally devised by botanists (Clark and Evans 1954) seeking to describe plant distribution is now widely used by geographers and is called the „nearest neighbor index“

16.1 The Nearest Neighbor Index

This is a technique that deals with the patterns formed by points in an area. The importance of nearest neighbour index is that it shows, on continuous scale, comparison to be made of two or more spatial distributions. The technique is normally applied in the spatial description of settlement patterns but can also be applied to many distributions, such as retail and other services in an area. Theoretically, the nearest neighbour statistics (R_n) varies from 0.00 (indicating complete aggregation of points) through 1.00 (random distribution of points) to 2.149 (a perfectly regular arrangement of points in a hexagonal pattern). Another important value is 0.23, which indicates a linear arrangement of points as shown in figure 16.1. *Fig 16.1 The R_n Scale*

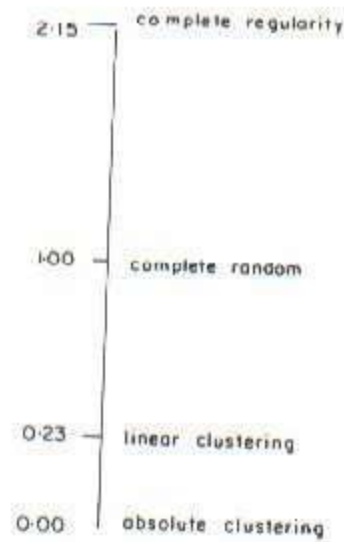


Fig. 16.1

The nearest neighbor statistics (R_n) is defined by the formula

$$R_n = \bar{\rho}_{rb} / \bar{\rho}_{re} = \bar{\rho}_{rb} / 2\sqrt{n/A}$$

Where $\bar{\rho}_{rb}$ is observed mean nearest neighbor distance between the points; $\bar{\rho}_{re}$ is the expected mean nearest neighbor distance between points assuming a random distribution in accordance with a Poisson probability function; n is the number of points in the study region, and A is area of study region.

16.1.1 Procedure Followed in the Calculation of Nearest Neighbor Statistics (R_n)

Calculation of nearest neighbor coefficient (R_n) involves six steps:

Step 1: Delimit the region and ascertain the actual area (A) of the region containing the points.

Step 2: Calculate the density of points in the region. The density (p) is obtained by dividing the number of points (n) in the region by the area of the region (A) i.e.,

$$P = \text{number of points in study region} / \text{Area of study region}$$

$$P = n/A$$

Step 3: Measure the nearest neighbor distance for each point. To do this, each point is taken individually and the distance to the nearest point is measured. This is done for all the points in the region (A), and measure the distances in the same unit as used in the calculation of A . Note that when two points are close together, they are each others nearest neighbor, and the nearest distances will be the same for both. These points, which are closer to each other than to any other point, are called reflexive pairs.

Step 4: Calculate the observed mean of the nearest neighbor distance ($\bar{\rho}_{rb}$) by adding up all nearest neighbor distances and divide by the number of points (n).

Step5: Calculate the expected mean nearest neighbor distance ($\bar{\rho}_{re}$) between the points if the pattern assumed a random distribution in occurrence with a Poisson probability function using the formula:

$$\bar{\rho}_{re} = 1/2\lambda P$$

Where $\bar{\rho}_{re}$ is the expected nearest neighbor distance, if the distance was random.

Step 6: The final stage is to calculate the nearest neighbor statistic (RR), which is the ratio of $\bar{\rho}_{ro}$ to $\bar{\rho}_{re}$, that is $R_n = \bar{\rho}_{ro}/\bar{\rho}_{re}$

Where $\bar{\rho}_{ro}$ is the observed mean nearest neighbor distance, $\bar{\rho}_{re}$ is the expected mean nearest neighbor, if the pattern was random.

16.1.2 Significance of the R_n Value

1. If R_n value indicates a random distribution, it may be that the pattern results from forces working at random, but no firm conclusion should be drawn from this result. Whilst the method may be used to prove that the distribution is random, it certainly cannot be used to prove what it is.
2. Of greater value is the indication of tendency towards clustering as dispersion for which the underlying causes may be sought. Nearest neighbor analysis is therefore useful in isolating and defining a problem.
3. Interpretation will also depend on the nature of the problem being investigated. The statistic reveals certain tendencies enabling formulated hypotheses to be accepted or rejected.
4. It enables more exact comparisons to be made, i.e., it demonstrates clearly the point patterns that are not random, which therefore require some explanations given that phenomena are supposed to be randomly distributed in nature.
5. It “averages out” sub-patterns which may exist within the area, and thereby hide contrasting sub-patterns which cancel each other out when put together to give a false

impression of randomness. For example, it does not distinguish between a single and multi-clustered pattern.

6. The selection of the first nearest neighbor is arbitrary and time saving. The choice of the second and third nearest neighbor is possible and might produce different results.
7. The fact that a pattern emerges with an index approaching 1.0, does not mean that the location of the observed phenomena in the real landscape is necessarily purely the result of 'chance'. The distribution of one kind of phenomena may largely determine that of another. For example, population distribution may greatly affect the location of the market places, as a particular type of drift deposited by melting ice in an apparently random manner might closely affect the pattern of agricultural land use.
8. There is also the problem of boundary.

16.2 Exercise

1. Given that in a certain region of area 7,151,000 m², there are 7 churches labeled A to G. The distances of the churches to the nearest neighbor are in the Table 16.0.

Table 16.0: The Distance of the churches to the nearest neighbor in metres

<i>Church</i>	<i>Distance</i>
A	150
B	150
C	130
D	240
E	130
F	230
G	230
Sum total	1260

Calculate the nearest neighbor index (R_n) and comment on the results.

1. A settlement geographer delimited a region 17.3 km long and 13.2 km wide. From a map, he identified 27 villages in the region and numbered them point 1 to 27. He also measured the nearest neighbor distances between the villages and came up with table 16.1.

Table 16.1: An area measuring 17.3km by 13.2 km with 27 villages

BY ODHIAMBO DARREN GREGORY

+254704597085

Village	Nearest Neighbor Village	Nearest Neighbor Distance (NND)
1	2	2.50
2	3	1.60
3	2	1.60
4	5	2.10
5	7	1.00
6	9	1.45
7	5	1.00
8	7	1.55
9	6	1.45
10	11	0.80
11	10	0.80
12	14	1.00
13	14	0.95
13	14	0.95
14	13	0.95
15	14	0.75
16	15	1.25
17	16	1.30
18	21	2.10
19	20	3.00
20	19	3.00
21	18	2.10
22	18	4.15
23	24	2.55
24	23	2.55

25	24	1.50
26	23	2.00
27	26	2.35
□NND		47.35

Use the nearest neighbor index to describe the spatial distribution of the villages.

SECTION 17

17.0 NETWORK ANALYSIS IN ECONOMIC GEOGRAPHY 17.1 Introduction

The major use of network analysis in economic geography concerns transport networks. Transport routes do not exist all by themselves, but they are organized into a network. Individual networks such as shipping lines, railroads, highways and pipeline service transport, demand and bind regions together.

17.2 Network as Graphs

Networks are enormously complex spatial system each with its own bundle of characteristics. This makes them difficult to describe, evaluate and compare. One approach that examines the basic elements of networks and permits us to explore their common classical characteristics is provided by graph theory.

Network analysis is based on the transformation of a transportation network to a graphical form. This simplifies the network into nodes as vertices (V), edges (e) and sub-graphs (p) and disregards other properties such as distances, mode sizes, orientation, traffic flow, etc. Also omitted at this stage are factors that influence the development of the network such region-shape and size-relief of the land and population distribution. The analysis uses the graph theory, a topology to devise measures, which describe the network as a graph. K.J. Kamsky (1963) devised this method of analysis and it enables us to: (a) Express exact relationships between network properties,

(b) Establish an order of networks and particular properties of that network,

- (c) Use graphs as simple models of the real world to give a visual impression of spatial relationships.

17.2.1 Procedures Followed in Network Analysis

The first step in the analysis is to reduce the network to a graph made up of modes, edges and sub-graphs. The edges are the routes as lines joining places. They are often referred to as links, branches, or routes, but will therefore be known as edges and given the notation e . The nodes or vertices are places joined by the edges. They are given the notation V and should occur wherever edges intersect and at the end of edges. The sub-graphs are the number of graphs and networks under consideration as unconnected parts of the whole graph or network. They are generally denoted by p .

Transformation of a network into a graph

Figure 17.0(a) shows a simple network of roads connecting nine places. Figure 17.0(b) shows the same network simplified into a graph. The values of the element of the graph, e , v and p are given within the diagram. Such a graph shows topographical position only. The location of nodes is considered in terms of relative position of the graph regardless of their absolute location in the real world. Distances between nodes are determined in terms of intervals, not route length.

Using the characteristics of vertices and edges, one can estimate the efficiency of a network by means of accessibility indices that measure the effort necessary to reach certain vertices of the network from one vertex, and in terms of connectivity indices that evaluate the effort necessary to reach all vertices from all others.

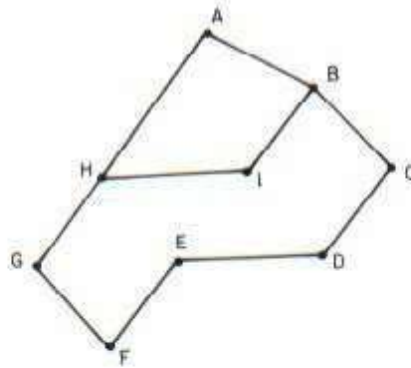


Figure 17.0(a):A Simple Network Connecting Places

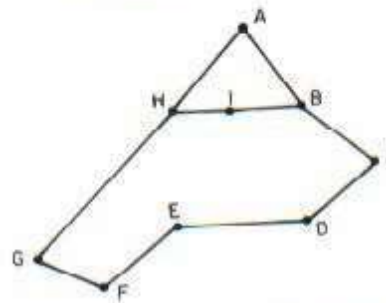


Figure 17.0(b): Network in Fig.17.0(a) Simplified into a Graph

17.3 Network Connectivity

A number of graphic theoretical measures namely the data index; the gamma index, the alpha index and the cyclomatic number may be used to evaluate the connectivity of networks.

Beta index (β) expresses the ratio between the number of edges (e) in a system and the number of vertices (v) in that system $\beta = e/v$

For example in figure 17.0(b), the beta index, $\beta = 10/9 = 1.1$, where the number of edges to vertices is large, the beta value will also be large, indicating a well-connected network.

Conversely, more vertices than edges, signifies a poorly connected network.

The beta index may be used to compare the structure of the networks, either several different networks at the same time or single network as it changes through time. Beta values are generally high in underdeveloped areas

Gamma Index (γ)

This is another graphic theoretical measure of connectivity. The γ is given by the formula:

$$\gamma = 100 \cdot e / 3(v-2) \quad \text{Equation 1,}$$

Whose values range from 0% for a non-interconnected network to 100% indicating a network that is completely connected.

Considering equation 1, the gamma index whose $e = 10$ and $v = 9$ can be computed as follows:

$$\begin{aligned} \gamma &= 100 \cdot 10 / 3(9-2) \\ &= 100 \times 10 / 21 \\ &= 47.6\% \end{aligned}$$

The γ is independent of the number of nodes in the network and can therefore be used to compare networks of different sizes. It is always advisable to employ more than one index of connectivity when studying networks. The cyclomatic number and the alpha index incorporate more than just the number of edges and nodes in the analysis, as they can also use the idea of circuits in a network.

The cyclomatic number

It is generally denoted by (β), which gives the number of circuits in the network or graph. It is given by the formula:

$$\beta = e - v + p \quad \text{Equation 2}$$

Again considering, say the values of $e = 10$, $v = 9$, $p = 1$; the β can be calculated as follows:

$$\begin{aligned} \beta &= 10 - 9 + 1 \\ &= 2 \end{aligned}$$

A high value of β indicates a highly connected network. It has been shown that there exists a significant relationship between a region's level of economic development and the value of the β its major transport networks.

Alpha Index

Letter denotes alpha index α , which is the ratio of the actual number to the maximum possible number of circuits in the network. It relates the actual cyclomatic number to the maximum possible cyclomatic number. The formula for the index is shown by equation 3:

$$\alpha = 100 \cdot \frac{e - v + p}{2v - 5} = 100 \cdot \frac{\alpha}{2v - 5} \quad \text{Equation 3}$$

Considering the example (figure b), the $\alpha = 100 (10 - 9 + 1) / (2 \times 9 - 5)$

$$= 2/3 \times 100 = 15.38$$

As with gamma index, values range from 0% to 100%, the latter indicating a completely interconnected network and the former indicating a “tree” type of network.

□ One common problem to measures of network connectivity is that they have low discriminatory power, i.e., they are unable to discriminate between networks having the same number of edges and vertices. Employing these simple measures, insight is gained into studies of transport and communications, which might at first seemed too complex to tackle. Several geographers have shown that measures of national development such as per capita gross national product (GNP), energy consumption and composition of exports increases so does the values of α , β , and γ . The reason behind this correlation is that the advanced economies have highly integrated road, rail, and air networks. The measures described here may be seen as a surrogate of the level of economic development when used at national scale.

17.4 Exercise

Figure 17.1 below, shows a motorway network of a certain economic planning unit.

Using the network, calculate the following graphic theoretical measures of connectivity:

- (a) α Index, □□
- (b) Gamma Index, □□

(c) Cyclomatic Number, $\square\square$ (d) Alpha Index, $\square$.

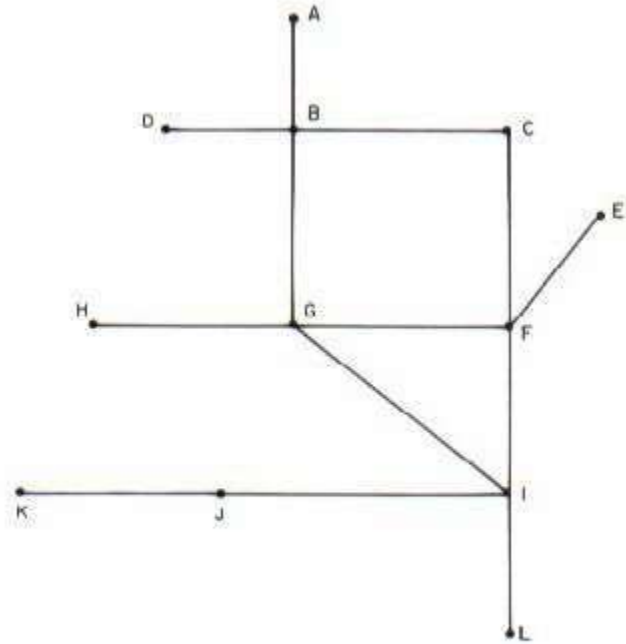


Fig.17-1: A Motorway Network of a Hypothetical Planning Region

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